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PHYSICS

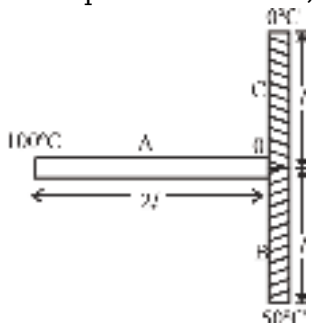
1) A gas has molar heat capacity C = 4.5 R in the process PT = constant. The number of degrees of freedom of molecules in the gas is :-

(1) 4
(2) 3
(3) 6
(4) 5

2) 10 gm of ice at –20°C is dropped into a calorimeter containing 10 gm of water at 10°C; the specific heat of water is twice that of ice. When equilibrium is reached, the calorimeter will contain :-

(1) 20 gm of water
(2) 20 gm of ice
(3) 10 gm ice and 10 gm of water
(4) 5 gm ice and 15 gm of water

3) Three rods A, B and C of thermal conductivities K, 2K and 4K, cross-sectional areas A, 2A and 2A and lengths 2l, l and l respectively are connected as shown in the figure. If the ends of the rods are maintained at temperatures 100°C, 50°C, and 0°C respectively, then the temperature θ of the



junction is:-

(1) $\frac{300}{7}$C
(2) 20°C
(3) $\frac{200}{7}$C
(4) $\frac{200}{13}$C

4) An ideal gas follows VT^2 = const. Its volume expansion coefficient will be :-

(1) $\frac{2}{T}$

(2) $-\frac{2}{T}$

(3) $\frac{3}{T}$

(4) $-\frac{3}{T}$

5) The root mean square velocity of the molecules in a sample of helium is $\frac{1}{\sqrt{2}}$ times of that of the molecules in a sample of hydrogen. If the temperature of the hydrogen sample is 0°C, that of helium samples is approximately is about :-

(1) 0°C

(2) 0 K

(3) 273°C

(4) 100°C

6) 32 gm of O_2 gas is contained in a flask at a pressure 10 atm. and temperature 57°C. It is found that due to leakage in the flask, the pressure is reduced to half and temperature reduced to 27°C. The quantity of O_2 gas that leaked out :-

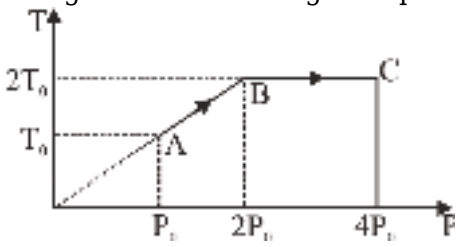
(1) 88/5 gm

(2) 72/5 gm

(3) 63/5 gm

(4) None

7) One mole of an ideal gas is taken through the process ABC as shown in the figure. The total work done on the gas is:

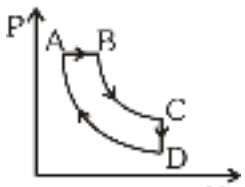


(1) zero

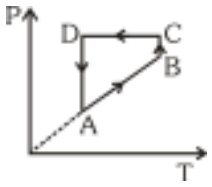
(2) $2RT_0 \square n2$

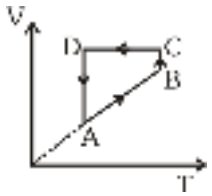
(3) $-2RT_0 \square n2$

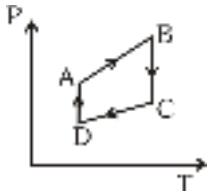
(4) $4RT_0 \square n2$

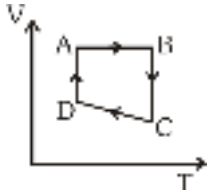


8) A cyclic process ABCDA is shown in the P–V diagram. Which of the following curves represent the same process? (Given BC & DA are isothermal curves)

(1) 

(2) 

(3) 

(4) 

9) A hot black body emits energy at the rate of 4 J $m^{-2}s^{-1}$ and its most intense radiation corresponds to 20,000 Å. When the temperature of this body is further increased and its most intense radiation corresponds to 10,000 Å then, the energy radiated in J $m^{-2}sec^{-1}$ will be :-

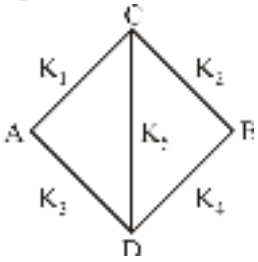
(1) 4

(2) 1

(3) 64

(4) 256

10) Five rods of same dimensions are arranged as shown. They have thermal conductivity K_1, K_2, K_3, K_4 and K_5. When points A and B are maintained at different temperatures, no heat flows through the central rod if :-



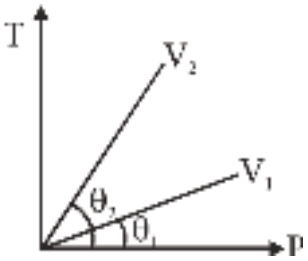
(1) $K_1K_4 = K_2K_3$

(2) $K_1 = K_4$ and $K_2 = K_3$

(3) $\frac{K_1}{K_4} = \frac{K_2}{K_3}$

(4) $K_1K_2 = K_3K_4$

11) From the following P-T graph, what inference can be drawn ?



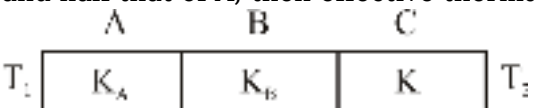
(1) $V_2 > V_1$

(2) $V_2 < V_1$

(3) $V_2 = V_1$

(4) None of the above

12) Three identical metal rods in dimension A, B and C are placed end to end and a temperature difference is maintained between the ends of A and C. If thermal conductivity of B is thrice that of C and half that of A, then effective thermal conductivity of the system is :-



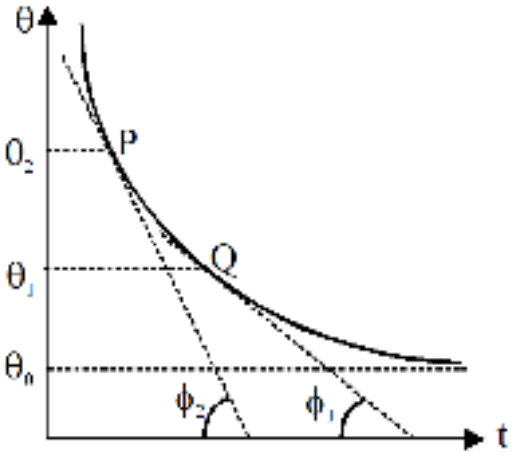
(1) 12K/7

(2) 6K/7

(3) 7K/3

(4) 2K

13) A body cools in a surrounding which is at a constant temperature of θ_0. Assume that it obeys Newton's law of cooling. Its temperature θ is plotted against time t. Tangents are drawn to the curve at the points $P(\theta = \theta_2)$ and $Q(\theta = \theta_1)$. These tangents meet the time axis at angels of ϕ_2 and ϕ_1, as shown



(1) $\frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_1 - \theta_0}{\theta_2 - \theta_0}$

(2) $\frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$

(3) $\frac{\tan\phi_1}{\tan\phi_2} = \frac{\theta_1}{\theta_2}$

(4) $\frac{\tan\phi_1}{\tan\phi_2} = \frac{\theta_2}{\theta_1}$

14) If an electric heater is rated at 1000 W, then the time required to heat one litre of water from 20°C to 60°C is :-

(1) 1 min 24 sec

(2) 2 min 48 sec

(3) 4 min 17 sec

(4) 5 min 36 sec

15) Outside atmospheric temperature of Dal lake in Srinagar is –θ°C. If density of water is ρ and its specific heat capacity is S and thickness of frozen ice is present at the lake already. Find rate of increase of thickness of ice at this instant. If k is thermal conductivity of ice and L is specific latent heat of fusion of ice.

(1) $\frac{k\theta}{\rho \ell L}$

(2) $\frac{\rho \ell}{Lk\theta}$

(3) $\frac{\rho \ell L}{k\theta}$

(4) $\frac{Lk\theta}{\rho \ell}$

16) The amount of heat energy required to raise temperature of 1 g of helium at constant volume from T_1 K to T_2 K is

(1) $\frac{3}{4}N_A k_B(T_2 - T_1)$

(2) $\frac{2}{3}N_A k_B(T_2 - T_1)$

(3) $\frac{3}{2}N_A k_B(T_2 - T_1)$

(4) $\frac{3}{8}N_A k_B(T_2 - T_1)$

17) The temperature of a pendulum, the time period of which is t, is raised by ΔT. The change in its time period is:

(1) $\frac{1}{2}\alpha t \Delta T$

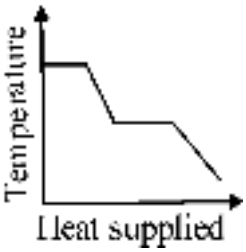
(2) $2\alpha t \Delta T$

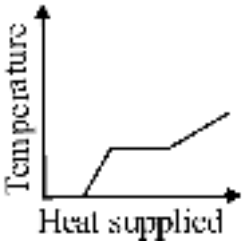
(3) $\frac{1}{2}\alpha \Delta T$

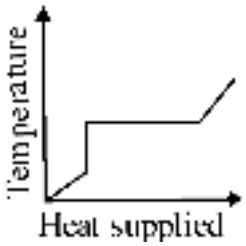
(4) $2\alpha \Delta T$

18) A block of ice at –100°C is slowly heated and converted to steam at 100°C. Which of the following curves represents the phenomenon qualitative?

(1) 

(2) 

(3) 

(4) 

19) If for hydrogen $C_P - C_v = m$ and for nitrogen $C_p - C_v = n$, where C_P and C_v refer to specific heats per unit mass relation between m and n is : (Molecular weight of hydrogen = 2 and molecular weight of nitrogen = 14)

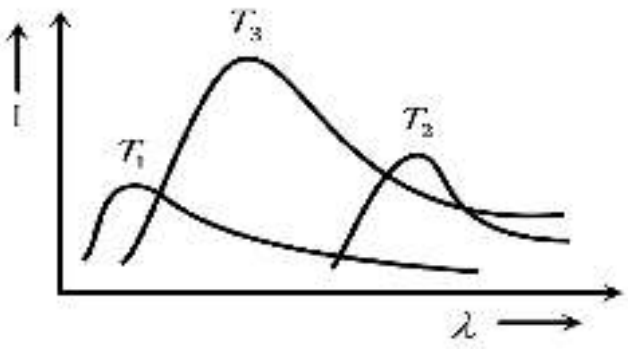
(1) n = 14m

(2) n = 7m

(3) m = 7n

(4) m = 14n

20) The plots of intensity versus wavelength for three black bodies at temperatures T_1, T_2 and T_3 respectively are as shown. Their temperature are such that



(1) $T_1 > T_2 > T_3$

(2) $T_1 > T_3 > T_2$

(3) $T_2 > T_3 > T_1$

(4) $T_3 > T_2 > T_1$

21) Two simple harmonic motions are represented by the equations :

$Y_1 = 10\sin\left(3\pi t + \frac{\pi}{4}\right)$ and

$Y_2 = 5\left(\sin 3\pi t + \sqrt{3}\cos 3\pi t\right)$.

Their amplitudes are in the ratio of :-

(1) 2 : 1

(2) 3 : 1

(3) 1 : 4

(4) None of the above

22) If equation of SHM are given as $x_1 = 4\sin\omega t$ and $x_2 = 4\cos\omega t$ when both SHM are superimposed then we will get resultant SHM then find total energy in resultant SHM (If K = 100 N/m for both the equation)

(1) 1600 J

(2) 800 J

(3) 400 J

(4) 200 J

23) When the cart maximally compresses the spring at the bottom of the fixed track, the cart's



(1) velocity and acceleration are zero

(2) velocity is nonzero but its acceleration is zero

(3) acceleration is nonzero, but its velocity is zero

(4) velocity and acceleration are both nonzero

24) A particle executes simple hormonic motion between x = –A and x = +A. It starts from x = 0 moves in +x-direction. The time taken for it to move from x = 0 to x = $\frac{A}{2}$ is T_1 and to move from $\frac{A}{2}$ to $\frac{A}{\sqrt{2}}$ is T_2, then

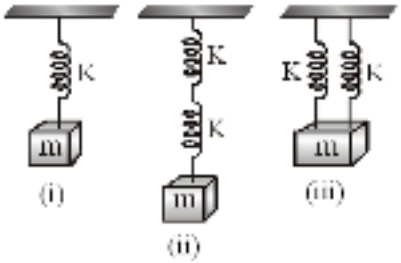
(1) $T_1 < T_2$

(2) $T_1 = T_2$

(3) $T_2 = 2T_1$

(4) $T_1 = 2T_2$

25) Five identical springs are used in the following three configurations. The time periods of vertical



oscillations in configurations (i), (ii) and (iii) are in the ratio:-

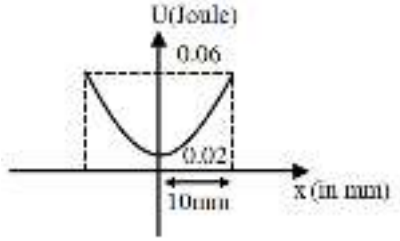
(1) $1 : \sqrt{2} : \frac{1}{\sqrt{2}}$

(2) $2 : \sqrt{2} : \frac{1}{\sqrt{2}}$

(3) $\frac{1}{\sqrt{2}} : 2 : 1$

(4) $2 : \frac{1}{\sqrt{2}} : 1$

26) The variation of potential energy of harmonic oscillator is as shown in figure, the spring constant is :-



(1) 200 N/m

(2) 600 N/m

(3) 800 N/m

(4) 400 N/m

27) Four massless springs whose force constants are 3K, 6K, K and 2K respectively are attached to a mass M Kept on a friction less plane as shown in the figure. If the mass M is slightly displaced in horizontal direction, then the time period of oscillation of the system is :-



(1) $2\pi\sqrt{\frac{M}{K}}$

(2) $2\pi\sqrt{\frac{M}{12K}}$

(3) $2\pi\sqrt{\frac{M}{6K}}$

(4) $2\pi\sqrt{\frac{M}{5K}}$

28) A simple pendulum is executing simple harmonic motion with a time period T. If the length of the pendulum is increased by 96%, the increase in the time period of the pendulum is :-

(1) 48%

(2) 44%

(3) 40%

(4) 33%

29) ***Assertion :*** Pendulum clocks go slow in summer and fast in winter.
Reason : The length of the pendulum used in clock increases in summer.

(1) Both Assertion & Reason are true & the reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are true but reason is not a correct explanation of the Assertion.

(3) Assertion is true but the Reason is false

(4) Both Assertion & Reason are false

30) For a particle executing SHM along a straight line, match the statements in column I with statement in column II. (Note that displacement given is column I is to be measured from mean position)

	Column I		**Column II**
i.	Velocity-displacement graph may be	a.	Straight line
ii.	Acceleration-velocity graph may be	b.	Circle
iii.	Acceleration-displacement graph will be	c.	Ellipse
iv.	Acceleration-time graph will be	d.	Sinusoidal curve

(1) i → a; ii → c, b; iii → b; iv → a

(2) i → b, c ; ii → b, c; iii → a; iv → d

(3) i → d, c ; ii → a, c; iii → d; iv → c

(4) i → a ; ii → d; iii → b; iv → c

31) The displacement of a particle executing simple harmonic motion is given by $y = A_0 + A\sin\omega t + B\cos\omega t$. Then the amplitude of its oscillation is given by

(1) A + B

(2) $A_0 + \sqrt{A^2 + B^2}$

(3) $\sqrt{A^2 + B^2}$

(4) $\sqrt{A_0^2 + (A + B)^2}$

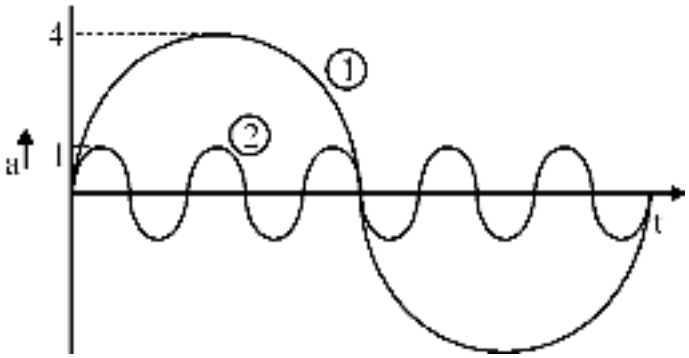
32) The path difference between the two waves $y_1 = a_1 \sin\left(\omega t - \frac{2\pi x}{\lambda}\right)$ and $y_2 = a_2 \cos\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right)$ is :-

(1) $\frac{\lambda}{2\pi}\phi$

(2) $\frac{\lambda}{2\pi}(\phi + \pi/2)$

(3) $\frac{2\pi}{\lambda}(\phi - \pi/2)$

(4) $\frac{2\pi}{\lambda}\phi$



33) For a given medium find intensity ratio I_1:I_2 :-

(1) 4 : 5

(2) 5 : 4

(3) 16:25

(4) 25:16

34) A string with a mass density of 4×10^{-3} kg/m is under tension of 360 N and is fixed at both ends. One of its resonance frequencies is 375 Hz. The next higher resonance frequency is 450 Hz. The mass of the string is -

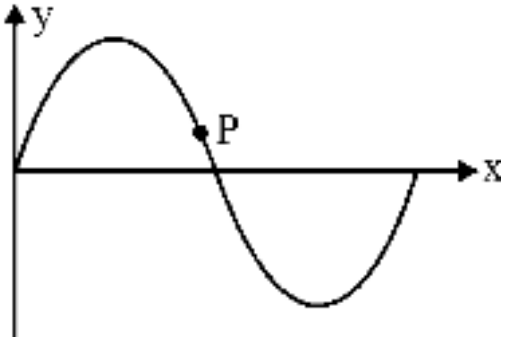
(1) 2×10^{-3} kg

(2) 3×10^{-3} kg

(3) 4×10^{-3} kg

(4) 8×10^{-3} kg

35) A transverse sinusoidal wave moves along a string in the positive x-direction at a speed of 10 cm/s. The wavelength of the wave is 0.5m and its amplitude is 10 cm. At a particular time t, the snap-shot of the wave is shown in fig. The velocity of P when its displacement is 5 cm is :-



(1) $\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s

(2) $-\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s

(3) $\frac{\sqrt{3}\pi}{50}\hat{j}$ m/s

(4) $-\frac{\sqrt{3}\pi}{50}\hat{i}$ m/s

36)

If the speed of sound at 0°C is υ_0 , then the velocity at 273°C will be :-

(1) $2\upsilon_0$

(2) $0.5\ \upsilon_0$

(3) $1.4\ v_0$

(4) $0.7\ \upsilon_0$

37) Find the speed of sound in a mixture of 1 mole of Helium and 1 mole of Hydrogen at 127°C :-

(1) 1290 m/sec

(2) 460 m/sec

(3) 620 m/sec

(4) 910 m/sec

38) A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a sonometer. The beat frequency decreases to 2 beats per second when the tension in the sonometer string is slightly increased. The frequency of the sonometer string before increasing the tension was :

(1) 510 Hz

(2) 514 Hz

(3) 516 Hz

(4) 508 Hz

39)

A glass tube of 100 cm length is filled with water. The water can be drained out slowly at the bottom of the tube. If a vibrating tuning fork of frequency 500 Hz is brought at the upper end of the tube and the velocity of sound in air is 330 m/s, then the total number of resonances obtained will be

(1) 4

(2) 1

(3) 2

(4) 3

40) Two vibrating tuning forks produce progressive waves given by $y_1 = 4\sin(500\pi t)$ and $y_2 = 2\sin(506\pi t)$. These tuning forks are held near the ear of a person. The person will hear α beats/s with intensity ratio between maxima and minima equal to β. Find the value of $\beta-\alpha$.

(1) 3

(2) 6

(3) 9

(4) 12

41) When both wave superimpose completely then the energy is in the form of -



(1) Kinetic Energy

(2) Potential Energy

(3) Partial potential & partial K.E

(4) Energy will completelty Loss

42) 25 turning forks are arranged in series in the order of decreasing frequency. Any two successive forks produce 3 beats/sec. If the frequency of the first tuning fork is the octave of the last fork, then the frequency of the 21st fork is :

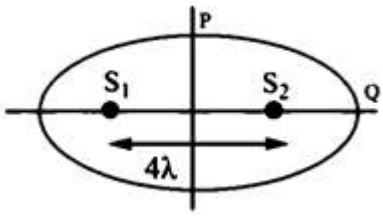
(1) 72 Hz

(2) 288 Hz

(3) 84 Hz

(4) 87 Hz

43) S_1, S_2 are two coherent sources (having initial phase difference zero) of sound located along x-axis separated by 4λ where λ is wavelength of sound emitted by them. Number of maxima located on the elliptical boundary around it will be : [S_1 and S_2 are assumed to be at focus of ellipse]



(1) 16

(2) 12

(3) 8

(4) 4

44) A person is talking in a small room and the sound intensity level is 60 dB everywhere within the room. If there are eight people talking simultaneously in the room, what is the sound intensity level ?

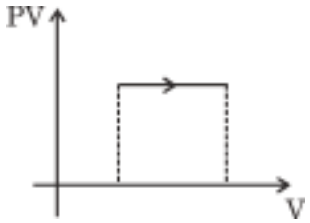
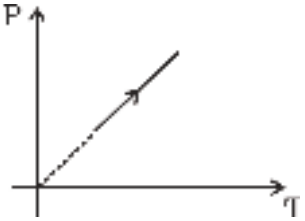
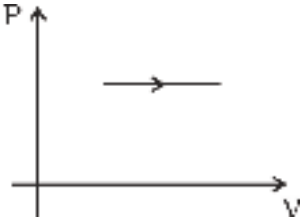
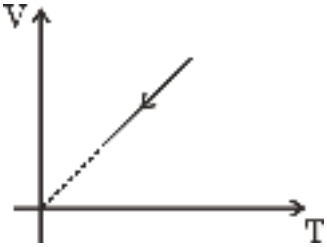
(1) 60 dB

(2) 69 dB

(3) 74 dB

(4) 81 dB

45) Column-I shows graph of thermodynamics process, and column-II contains information about various thermodynamic variables. Correct by match process in column-I with the information in column-II.

Column I		Column II	
(A)		(P)	W > 0
(B)		(Q)	W ≤ 0
(C)		(R)	ΔQ > 0
(D)		(S)	ΔU > 0
		(T)	ΔU ≤ 0

(1) A → PST, B → QR, C → PS, D → QT

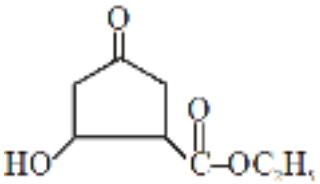
(2) A → PRT, B → QRS, C → PRS, D → QT

(3) A → PRT, B → QRS, C → PRS, D → QS

(4) A → PST, B → QR, C → PS, D → QT

CHEMISTRY

1) The correct IUPAC name of given compound is–



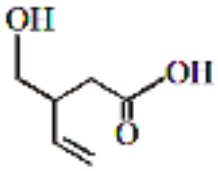
(1) Ethyl-3-hydroxy-5-oxo cyclo hexanoate

(2) Ethyl-2-hydroxy-4-oxo cyclo pentane carboxylate

(3) Ethyl-5-hydroxy-3-oxo cyclo pentane carboxylate

(4) Ethyl-6-hydroxy-4-oxo cyclo hexanoate

2) IUPAC name of given compound is :-



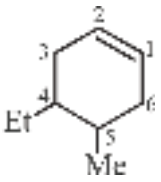
(1) 4–Hydroxy–3–vinylbutanoic acid

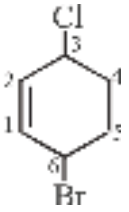
(2) 3–Methylhydroxypent–4–enoic acid

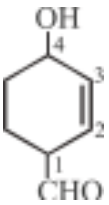
(3) 3–Hydroxymethylpent–4–enoic acid

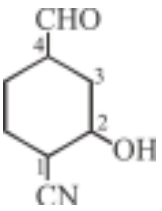
(4) 3–Hydroxymethylpent–4–enecarboxylic acid

3) In which case IUPAC numbering is incorrect ?

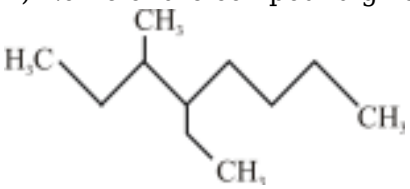
(1) 

(2) 

(3) 

(4) 

4) Name of the compound given below is :-

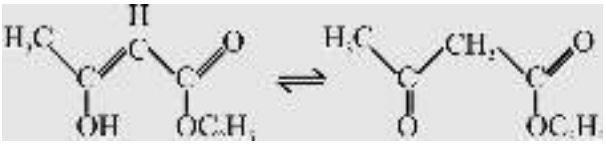


(1) 3-Methyl-4-Ethyloctane

(2) 4-Ethyl-3-methyloctane

(3) 2,3-Diethylheptane

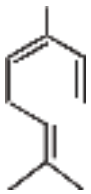
(4) 5-Ethyl-6-Methyloctane

5) The enolic form of ethyl acetoacetate as below has 

(1) 16 sigma bonds and 1 pi-bond
(2) 9 sigma bonds and 2 pi-bond
(3) 9 sigma bonds and 1 pi-bond
(4) 18 sigma bonds and 2 pi-bond

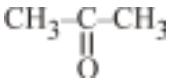
6) Correct name for given compound is : CH_2=CH–CN

(1) Vinyl cyanide
(2) Acrylonitrile
(3) Prop-2-ene-1-nitrile
(4) All of above

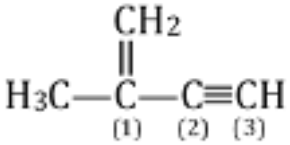
7) Which of the following is correct for the given compound :- 

(1) It is having 3 olefinic bonds
(2) It is a member of homologous series C_nH_{2n-6}
(3) Its IUPAC name is 3,7-dimethylocta–1,3,5–triene
(4) All

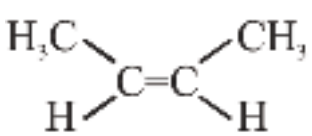
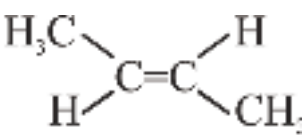
8) Which of following is not correctly matched?

(1) Common name of CH_2 = CH – OH is vinyl alcohol
(2) Common name of  is acetone
(3) Common name of CH_3–CH=CH–CHO is crotonaldehyde
(4) Common name of Ph – CH = CH–CHO is Acrylaldehyde

9) Hybridisation of C number (1), (2), (3) is:



(1) sp^2, sp, sp
(2) sp^3, sp^2, sp^2
(3) sp, sp^2, sp
(4) sp^2, sp^2, sp

10) 
(A)

(B)

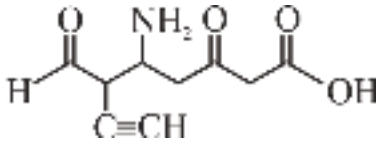
Which of the following statement is incorrect?

(1) A has higher boiling point than B.

(2) A has higher water solubility than B.

(3) B has higher melting point than A.

(4) B has lower stability than A.

11) 

Total number of functional group in given compound is/are :-

(1) 4

(2) 6

(3) 5

(4) 1

12)

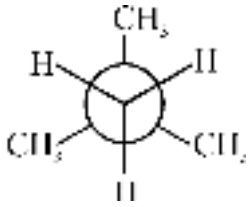
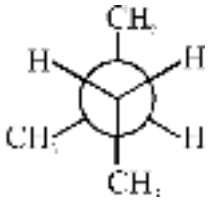
The number of total isomers of formula $C_2H_2Br_2$ are :-

(1) 2

(2) 3

(3) 4

(4) 1

13)  and  are :

(1) Conformational isomers

(2) Positional isomers

(3) Chain isomers

(4) Identical

14) Total number of isomer's having molecular formula C_4H_8 :-

(1) 2

(2) 3

(3) 4

(4) 6

15) Choose the correct statement ?

(1) Cyclobutane is a planar compound

(2) Geometrical isomers are interconvertable at room temperature

(3) Steric strain is same in both anti & Gauche form

(4) Structural isomers have different connectivity of groups

16) The minimum no of carbon atom requires for an aldehyde to show functional isomerism with ketone is :-

(1) 3

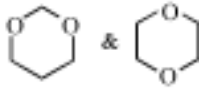
(2) 5

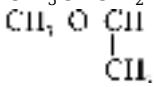
(3) 4

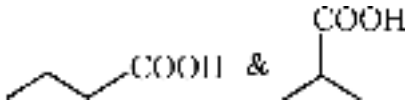
(4) 2

17) Which of following pairs of compounds are not position isomers ?

(1) 

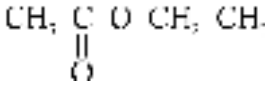
(2) 

(3) $CH_3O–CH_2–CH_2–CH_3$ &


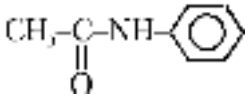
(4) 

18) Which does not show metamerism :-

(1) $CH_3–CH_2–O–CH_2–CH_3$

(2) 

(3) $CH_3–CH_2–NH–CH_3$

(4) 

19) Which of the following isomerisms is not found in alkenes ?

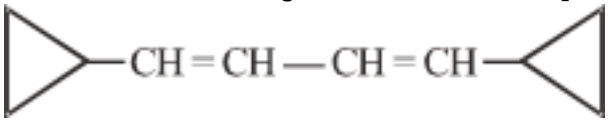
(1) Metamerism

(2) Geometrical isomerism

(3) Position isomerism

(4) Chain isomerism

20) Total number of geometrical isomers possible of the following compound



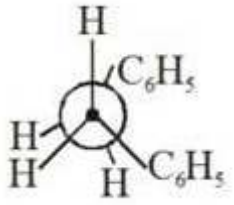
(1) 4

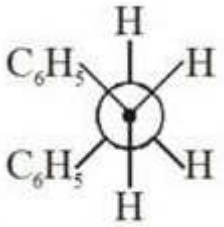
(2) 6

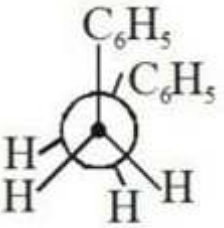
(3) 3

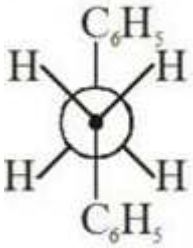
(4) 2

21) The most stable conformer of 1,2 - diphenyl ethane is:-

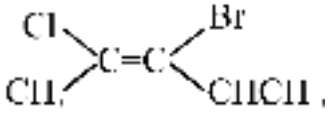
(1) 

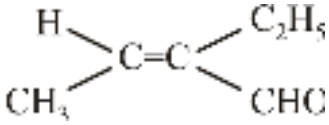
(2) 

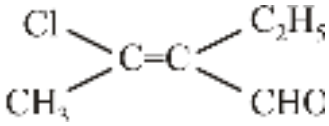
(3) 

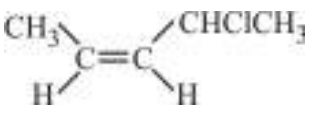
(4) 

22) Which among the following has E configuration?

(1) 

(2) 

(3) 

(4) 

23) The IUPAC name of the compound :- 

(1) (2E, 4Z, 6E)-2-Bromo-7-Chloro octa 2, 4, 6 triene
(2) (2Z, 4Z, 6E)-2-Bromo-7-Chloro-octa-2, 4, 6-triene
(3) (2E, 4E, 6Z)-2-Bromo-7-Chloro-octa-2, 4, 6-triene
(4) (2Z, 4Z, 6Z)-2-Bromo-7-Chloro-octa-2, 4, 6-triene

24) Which of the following compounds exhibit conformational isomerism ?

(1) But-2-ene
(2) Ethane-1,2-diol
(3) Maleic acid
(4) Fumaric acid

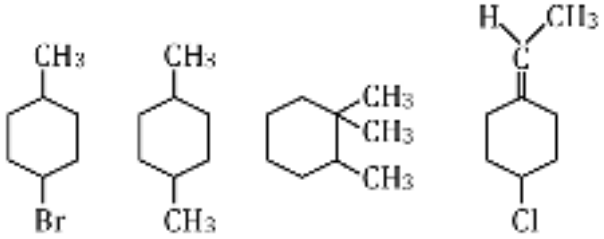
25) Which of the following shows geometrical isomers ?

(a) $CH_3—CH_2—CH=N—OH$ (b) $CH_2=N—OH$

(c) $CH_3—\underset{\underset{N—OH}{\|}}{C}—CH_3$ (d) $CH_3—\underset{\underset{N—OH}{\|}}{C}—CH_2—CH_3$

(1) a, c
(2) c, d
(3) a, d
(4) b, c

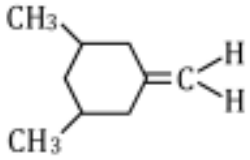
26) Which of the following compounds show geometrical isomers ?

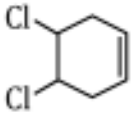


(1) I and II
(2) I and IV
(3) I, II and III
(4) All

27) Which of the following compounds will show geometrical isomerism ?

(1) $CH_3—CH_2—CH=N—OH$

(2) 

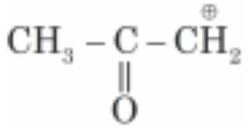
(3) 

(4) All of these

28) Stability order of following carbocation is :

(I) $CH_3-O-\overset{\oplus}{C}H_2$

(II) $CH_3-NH-\overset{\oplus}{C}H_2$

(III) 

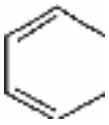
(1) II > I > III

(2) I > II > III

(3) III > I > II

(4) III > II > I

29) Which compound has most acidic H-atom ?

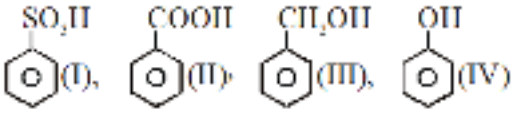
(1) 

(2) 

(3) 

(4) 

30) Arrange the following in correct order of acidic strength.

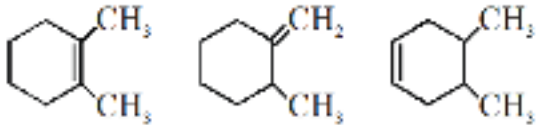


(1) I > II > III > IV

(2) I > II > IV > III

(3) II > I > III > IV

(4) II > III > IV > I

31)  (I) (II) (III)

Which of the following order is correct for stability :-

(1) I > II > III

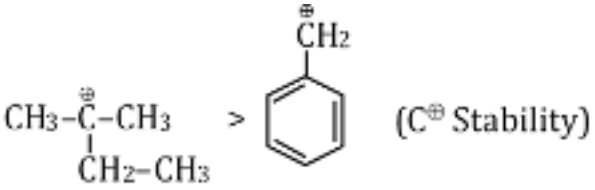
(2) III > II > I

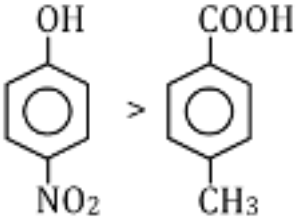
(3) I > III > II

(4) III > I > II

32) Which of the following statement correct :-

(1) $CH_3-\overset{\oplus}{C}H-CH_2-CH_3 > CH_3-\overset{\oplus}{C}H_2$ (Stability order)

(2)  ($C^{\oplus}$ Stability)

(3)  (Acidic Strength)

(4) $(C_2H_5)_3N > (C_2H_5)_2NH > C_2H_5 - NH_2$ (Basic strength)

33) The correct order of the basic strength of methyl substituted amines in aqueous solution is:

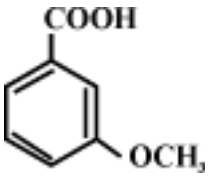
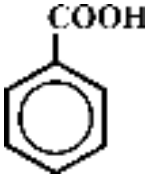
(1) $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N$

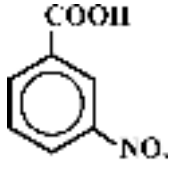
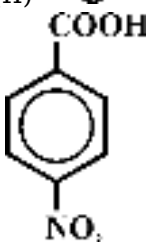
(2) $(CH_3)_3N > CH_3NH_2 > (CH_3)_2NH$

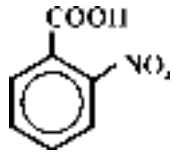
(3) $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2$

(4) $CH_3NH_2 > (CH_3)_2NH > (CH_3)_3N$

34) Find the correct order of k_a of following compounds.

(I)  (II) 

(III)  (IV) 

(V) 

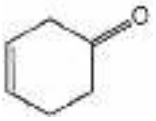
(1) I < II < III < IV < V

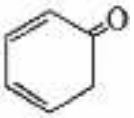
(2) IV < I < III < II < V

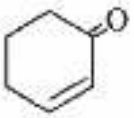
(3) III < II < I < IV < V

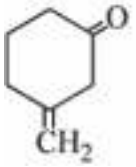
(4) II < I < III < IV < V

35) Which of the following has longest C–O bond

(1) 

(2) 

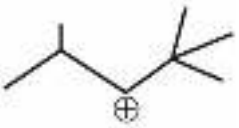
(3) 

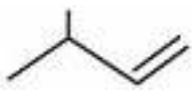
(4) 

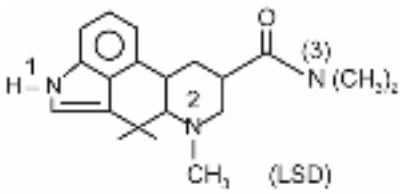
36) Hyperconjugation is not possible in:

(1) 

(2) 

(3) 

(4) 

37) 

(LSD) Which nitrogen in Lysergic acid diethylamide (LSD) is most basic?

(1) 1

(2) 2

(3) 3

(4) All are equally basic.

38) Which of the following will not act as an electrophile?

(1) B_2H_6

(2) $\overset{+}{N}O_2$

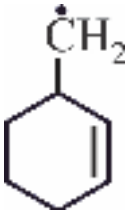
(3) SO_3

(4) $\overset{+}{N}H_4$

39) Decreasing order of stability for the following radicals is.

(I) $CH_2 = \dot{C}H$ (II) $\dot{C}H_2CH = CH_2$

(III) 

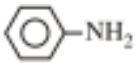
(IV) 

(1) II > III > I > IV

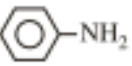
(2) III > II > IV > I

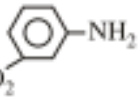
(3) III < II < I < IV

(4) I < IV < II < III

(I) 

(II) CH_3O 

(III) NO_2 

(IV) 

40)
The correct order of decreasing basicity of the above compounds is.

(1) I > II > III > IV

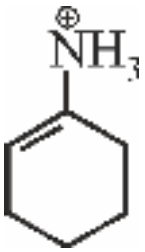
(2) II > I > IV > III

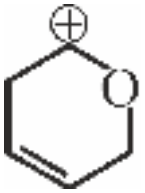
(3) III > IV > II > I

(4) II > I > III > IV

41) Which cation is involved in resonance ?

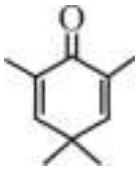
(1) $CH_2 = CH - \overset{\oplus}{O}H_2$

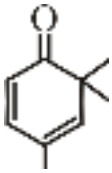
(2) 

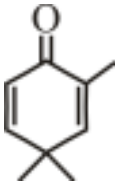
(3) 

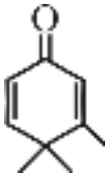
(4) All of these

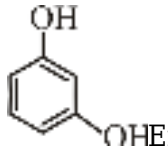
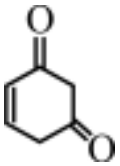
42) Tautomerism is shown by –

(1) 

(2) 

(3) 

(4) 

43) **Assertion :**  Enol is more stable than  Keto.
Reason : Enol is stabilised by aromaticity.

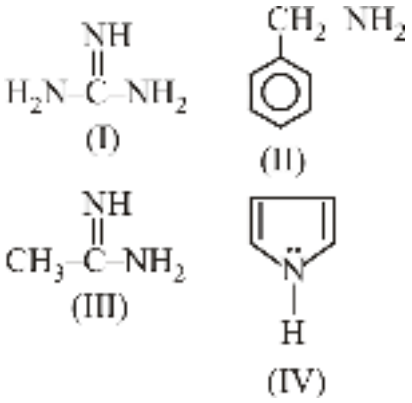
(1) Assertion and Reason are true but not correct explanation.

(2) Assertion is true but Reason is false.

(3) Assertion is false but Reason is true.

(4) Assertion and reason are true and correct explanation.

44) Correct order of basic strength



(1) I > II > III > IV

(2) I > III > II > IV

(3) I > IV > III > II

(4) I > III > IV > II

45) Match List I with List II.

List I Isomeric pairs		**List II Type of isomers**	
A.	Propanamine and N-Methylethanamine	I.	Metamers
B.	Hexan-2-one and Hexan-3-one	II.	Positional isomers
C.	Ethanamide and Hydroxyethanimine	III.	Functional isomers
D.	o-nitrophenol and p-nitrophenol	IV.	Tautomers

Choose the correct answer from the options given below :-

(1) A-III, B-IV, C-I, D-II

(2) A-IV, B-III, C-I, D-II

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-I, C-IV, D-II

BIOLOGY

1) How many statements are **correct** in following ?
(a) Net gain of 2ATP directly in glycolysis
(b) Final hydrogen acceptor in mitochondrial ETS is oxygen
(c) Glycolysis occurs in cytoplasm
(d) TCA cycle of prokaryotes occurs in cytoplasm

(1) One

(2) Two

(3) Three

(4) Four

2) No. of ATP molecules produced during complete oxidation of 1 pyruvate molecule would be :-

(1) 15
(2) 12
(3) 10
(4) 8

3)

NAD^+ of Krebs cycle functions as :-

(1) Acceptor of hydrogen ion and electrons
(2) Oxygen acceptor
(3) Oxygen donor
(4) Donor of phosphate ions

4)

Find the **incorrect** match:-

(1)	Complex-I	Oxidation of $NADH + H^+$
(2)	Complex-II	Oxidation of $FADH_2$
(3)	Complex-III	Metabolic water formation
(4)	Complex-V	ATP synthesis

(1) 1
(2) 2
(3) 3
(4) 4

5) Select the **wrong** statement.

(1) Plants do not have respiratory organs
(2) All organisms have the machinery to partially oxidise the glucose
(3) Respiratory substrates are oxidised in single step to release all energy contained in them
(4) Roots, stems and leaves respire at rates far lower than animals

6) The main purpose of cellular respiration is to:

(1) convert potential energy to kinetic energy
(2) convert kinetic energy to potential energy
(3) create energy in the cell
(4) convert energy stored in the chemical bonds of glucose to an energy that the cell can use

7) Which intermediate of Krebs cycle is used as a substrate for amino acid biosynthesis ?

(1) Citric acid
(2) α-ketoglutarate

(3) Succinyl-CoA
(4) All of the above

8) Select the **wrong** statement with respect to glycolysis:-

(1) It occurs outside mitochondria
(2) It is common for both anaerobic and aerobic respiration
(3) Glucose undergoes partial oxidation to form 2 molecules of pyruvic acid.
(4) Glucose is phosphorylated to glucose-6- phosphate by isomerase enzyme

9) Match List-I with List-II

	List-I		**List-II**
(A)	Oxidative decarboxylation	(i)	Cytoplasm
(B)	Glycolysis	(ii)	Inner mitochondrial membrane
(C)	Oxidative phosphorylation	(iii)	Pyruvate dehydrogenase
(D)	Tricarboxylic acid cycle	(iv)	H.A. Kreb

Choose the correct answer from the options given below :
(1) (A–i), (B–ii), (C–iii), (D–iv)
(2) (A–iii), (B–i), (C–iv), (D–ii)
(3) (A–iii), (B–i), (C–ii), (D–iv)
(4) (A–ii), (B–i), (C–iii), (D–iv)

10) All of the following are the assumptions made for making respiratory balance sheet, except

(1) There is sequential orderly pathway functioning with one substrate forming the next
(2) None of the intermediates in pathway are utilized to synthesise any other compound
(3) Only glucose is being respired
(4) Any substrate can enter the pathway at any of the intermediate stage

11) Select **incorrect** statement with respect to cytochrome C.

(1) It is present as small protein attached to outer surface of inner mitochondrial membrane
(2) It acts as mobile electron carrier
(3) It transfer electron between complex III and IV
(4) It is present as extrinsic protein on membrane cytochrome oxidase activity

12) Select the **wrong** pair :

(1) Succinate dehydrogenase: inner membrane of mitochondria
(2) F_0 – F_1 complex: oxysome
(3) Hexokinase matrix of mitochondria
(4) Pyruvate dehydrogenase complex: link reaction

13) How many $FADH_2$ molecules will be produced from 2 molecules of glucose in aerobic respiration?

(1) 4
(2) 2
(3) 1
(4) 8

14) **Assertion :-** Fermentation takes place under anaerobic condition in many prokaryotes and unicellular eukaryotes.
Reason :- During fermentation, incomplete oxidation of glucose takes place.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

15)

The metabolic fate of pyruvic acid depends on:-

(1) Availability of oxygen
(2) Availability of CO_2
(3) Availability of S
(4) Availability of N

16) Substrate level phosphorylation (SLP) occurs during:

(1) Link reaction and Krebs cycle
(2) Glycolysis and link reaction
(3) Krebs cycle and glycolysis
(4) Glycolysis only

17) **Statement-I :** The carbon skeleton produced during respiration is used as precursors for biosynthesis of other molecules in cell.
Statement-II : Plants, unlike animals, have no specialised organs for gaseous exchange but they have stomata and lenticels for this purpose.

(1) Only statement I is correct
(2) Only statement II is correct
(3) Both statement I and II are correct
(4) Both statement I and II are incorrect.

18) By aerobic respiration of 1 molecule of PGAL how many ATP get synthesized ?

(1) 8 ATP
(2) 2 ATP

(3) 19 or 20 ATP
(4) 36 ATP

19) Glycerol can enter the respiratory pathway after being converted into

(1) Pyruvic acid
(2) Phosphoglyceraldehyde
(3) Acetyl CoA
(4) 3-phosphoglyceric acid

20) When 2-pyruvic acids forms two lactic acid by anaerobic respiration then ?

(1) One ATP is lost
(2) 3 ATP is lost
(3) 6 ATP is lost
(4) 38 ATP are gained

21)

R. Q. is represented by –

(1) O_2/CO_2
(2) CO_2/ O_2
(3) H_2O/CO_2
(4) CO_2/H_2O

22) When tripalmitin is the respiratory substrate :-

(1) The amount of CO_2 released is less than the O_2 consumed
(2) The amount of CO_2 released is more than the O_2 consumed
(3) The amount of CO_2 released is equal to the O_2 consumed
(4) CO_2 is not released

23) Which of the following option is **correct** ?
(a) Respiration is oxidation of complex molecules to release energy (T/F)
(b) Respiration occurs in both cytoplasm and mitochondria (T/F)
(c) Respiration is only in eukaryotes but phososynthesis is in both prokaryotes and eukaryotes (T/F)
(d) Energy released during respiration is directly into cell (T/F)

(1) T T T T
(2) T T F F
(3) T F T F
(4) F F F F

24) Which of the following statements is incorrect :

(1) ATP is synthesised through complex V

(2) Oxidation - reduction reactions produce proton gradient in respiration

(3) During aerobic respiration, the role of oxygen is limited to the terminal stage

(4) In ETC [electron transport chain] one molecules of NADH + H^+ gives rise to 2ATP molecules and one FADH2 gives rise to 3ATP

25) Which of the following biomolecules is common to respiration-mediated breakdown of fats, carbohydrates and proteins?

(1) Glucose-6-phosphate

(2) Fructose 1,6-bisphosphate

(3) Pyruvic acid

(4) Acetyl Co-A

26) **Assertion A:** Plant growth is considered unlimited because of the presence of meristems, which have the ability to continuously divide and add new cells to the plant body. **Reason R:** Meristematic cells lose the capacity to divide after a certain number of divisions, leading to the cessation of growth.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

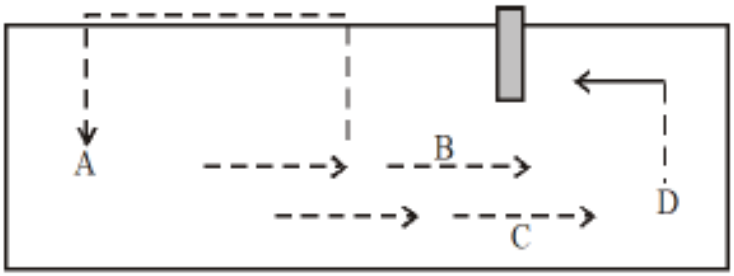
27) At cellular level, growth is :-

(1) increase in cytoplasm

(2) increase in size

(3) increase in nucleus size

(4) increase in protoplasm

28) 

Choose the correct option for A, B, C and D in given diagram :-

(1) A-mature cell, B-differentiation, C-maturation, D-meristematic cell

(2) A-meristematic cell, B-maturation C-differentiation, D-mature cell

(3) A-meristematic cell, B-differentiation, C-maturation, D-mature cell

(4) A-meristematic cell, B-differentiation, C-maturation, D-meristematic cell

29) One single maize root apical meristem can give rise to more than

(1) 17,500 new cells per minute

(2) 17,500 new cells per hour

(3) 17,500 new cells per day

(4) 17,500 new cells per week

30) Increased vacuolation, cell enlargement and new cell wall deposition are characteristic of cells in _____ phase of growth :-

(1) Meristematic

(2) Elongation

(3) Maturation

(4) Differentiation

31) $W_1 = w_0 e^{rt}$
r is not a :

(1) Relative growth rate

(2) The measure of the ability of the plant to produce new plant material

(3) Reffered as efficiency index

(4) Absolute growth rate

32) Match the PGR with the plants which played role in their discovery-

Column-I		**Column-II**	
(a)	Auxin	(i)	Tobacco
(b)	Gibberellin	(ii)	Rice
(c)	Cytokinin	(iii)	Orange
(d)	Ethylene	(iv)	*Avena*

(1) a-iii, b-i, c-ii, d-iv

(2) a-iv, b-ii, c-i, d-iii

(3) a-i, b-iii, c-iv, d-ii

(4) a-ii, b-iv, c-iii, d-i

33) Carotenoids, indole and terpenes are the precursors of phytohormone like _____ , _____ and _____ respectively.

(1) Auxin, GA and ABA

(2) Ethylene, ABA and GA

(3) Cytokinin, auxin and ABA

(4) ABA, auxin and GA

34) Which hormone is recognised for it's growth promoting as well as growth inhibiting role ?

(1) Ethylene

(2) Cytokinin

(3) ABA

(4) Auxin

35) How many of following functions are not related to Gibberellins ?
Internode elongation, malting, bolting, seed germination, abscission, potato dormancy, increasing yield of sugarcane, promote seed dormancy.

(1) 2

(2) 3

(3) 4

(4) 5

36) Which of the following is **not** a function of plant growth regulator that was first isolated from corn-kernels and coconut milk?

(1) Promotion of female flowers in monoecious plants.

(2) Production of chloroplast in leaves.

(3) Growth of lateral shoot.

(4) Overcome the apical dominance.

37) Which of the following plant growth regulations is **not** of acidic nature?

(1) Plant growth regulator which helps in overcome the apical dominance

(2) Plant growth regulator which initiates rooting in stem cutting

(3) Plant growth regulator which promotes bolting in rosette habit plants

(4) Plant growth regulator which stimulates the closure of stomata

38) Wrong pair is/are ....
(a) H.H. Causins - Ripened orange - Ethylene
(b) F.W. Went - Oat seedling - Indole compound.
(c) E.Kurosava - Fungus - Cytokinins
(d) F. Skoog and Miller - Herring fish sperm DNA - Kinetin

(1) a and b

(2) b and c

(3) Only d

(4) Only c

39) Read the given functions of X.
I. Helps in cell division.
II. Promote flowering in pineapple.
III. Induces parthenocarpy in some plants.
IV. Stimulate root formation on the stem cutting.
Here, 'X' is

(1) Gibberellin

(2) Auxin
(3) Cytokinin
(4) Ethylene

40) Seed dormancy :

(1) is the temporary delay to the process of germination.
(2) is the permanent delay to the process of germination.
(3) minimizes seedling survival by preventing germination.
(4) occurs due to the presence of growth inhibitors only.

41) **Assertion (A) :** Buttercup plant shows heterophilly
Reason (R) : Development of heterophylly is due to environment

(1) Both A and R are correct but R is not the correct explanation of A
(2) A is correct but R is not correct
(3) A is not correct but R is correct
(4) Both A and R are correct and R is the correct explanation of A

42) Which of the following is a energy yielding process ?

(1) 2-Phosphoglycerate to 2-phosphoenol pyruvate
(2) Fructose to Fructose-6-phosphate
(3) Phosphoenol pyruvic acid to pyruvic acid
(4) Fructose 1, 6-biphosphate to PGAL and DHAP

43)

In Krebs cycle FAD participates as electron acceptor during conversation of :-

(1) Succinyl CoA to succinic acid
(2) α-Keto glutarate to succinyl CoA
(3) Succinic acid to fumaric acid
(4) Fumaric acid to malic acid

44) Select correct option w.r.t. number of carbon present in respiratory intermediate :-

	Pyruvic acid	**Acetyl CoA**	**Citric acid**	**OAA**
(1)	3C	2C	5C	4C
(2)	2C	2C	6C	4C
(3)	3C	2C	6C	4C
(4)	3C	2C	6C	5C

(1) 1
(2) 2

(3) 3

(4) 4

45) How many molecule of CO_2 are released by Kreb's cycle during complete oxidation of one molecules of sucrose?

(1) 4

(2) 16

(3) 8

(4) 24

46) Which of the statement is **correct** about striated muscles ?

(1) They are found in urinary bladder and alimentary canal.

(2) Cytoplasm is known as exoplasm.

(3) It is syncytial or multinucleated.

(4) In sarcoplasmic reticulum potassium ions are stored.

47) How many bones are included in cranium :
Maxilla, Mandible, Sphenoid, Nasal, Ethmoid bone, Lacrymal, inferior turbinal, occipital bone :-

(1) 2

(2) 3

(3) 4

(4) 5

48) Which joint is similar to elbow joint ?

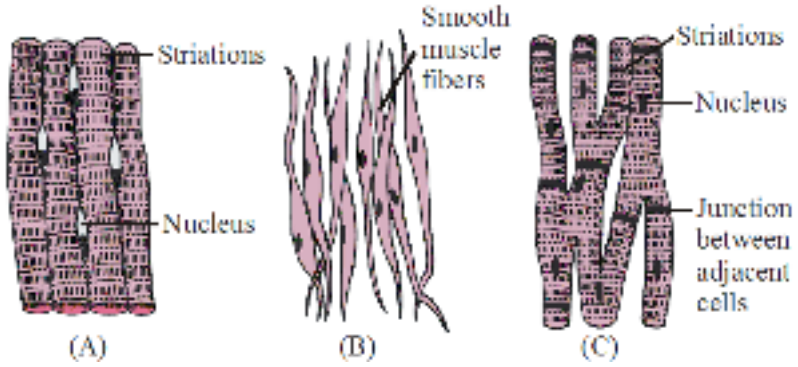
(1) Joint between humerus and scapula

(2) Joint between atlas and axis

(3) Joint between Femur, Tibia and patella

(4) Joint between Radius and Ulna

49) Select the **correct** match :-



(1) A-Skeletal muscles → biceps, striated muscles, involuntary

(2) B-Smooth muscles → walls of stomach, voluntary

(3) C-Cardiac muscles → heart, dorsal wall of aorta, voluntary

(4) A-Skeletal muscle → biceps, striated, voluntary

50) Choose the **incorrect** statement regarding actin structure :-

(1) Each actin (thin) filament is made of two F (filamentous) actins helically wound to each other.

(2) A complex protein troponin is distributed at regular intervals on the tropomyosin.

(3) Each F actin is a polymer of monomeric 'G' actin.

(4) In the resting state a subunits of troponin masks the active binding site for actin on the actin filament.

51)

How many of the following bones are belong to hind limb.

Patella, Tibia, Fibula, Ulna, Radius, Manubrium, Clavicle, Tarsals, Carpals, Metacarpal.

(1) 3

(2) 4

(3) 5

(4) 6

52) **Statement-I :-** Central part of myosin filaments, not overlapped by thin filaments is called the 'H' zone.
Statement-II :- When thin filaments slide over myosin filaments, 'H' zone becomes short and even may disappear during maximally contracted state.

(1) Both statement-I and statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but statement-II is correct.

(4) Both statement-I and statement-II are correct.

53) **Assertion (A) :-** Calcium plays an important role in skeletal muscle contraction.
Reason (R) :- Ca^{+2} binds to sub-unit of troponin to remove the masking of active sites for myosin on actin filaments.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

54) Main steps in the mechanism of muscle contraction are given below. Arrange these steps in a correct sequence.
(A) Shortening of sarcomere.
(B) Cross bridge formation.
(C) Release of Ca^{+2} ions into sarcoplasm.
(D) Removal of masking of active sites for myosin.
Choose the **correct** answer from the options given below :-

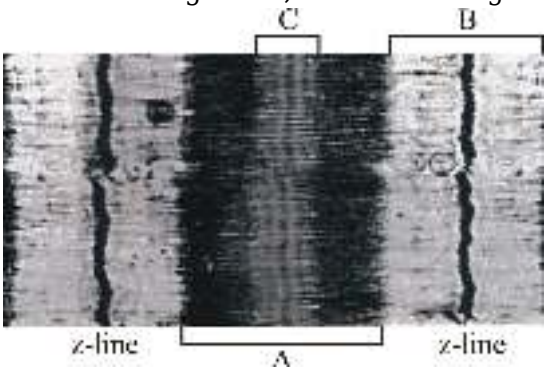
(1) C → A → B → D

(2) C → B → D → A

(3) D → C → A → B

(4) C → D → B → A

55) Choose the most appropriate option regarding length of labelled regions A, B and C during muscle contraction in the given structure of sarcomere :-



(1) B becomes short while A and C remain same.

(2) B and C become short while A remains same.

(3) A and C become short while B remains same.

(4) A becomes short while B and C remain same.

56) How many of the following statement are **correct** regarding muscle contraction ?
(a) A-band retain its length
(b) I-band gets reduced
(c) sarcomere gets shorter
(d) Actin & myosin filaments gets shorter
(e) Thick filament slide over thin filament

(1) Five

(2) Three

(3) Four

(4) Two

57) Select the **correct** option :-

(1) First vertebra is the axis and it articulates with the occipital condyles.

(2) Sternum is a flat bone on the dorsal midline of thorax.

(3) Thoracic vertebrae, ribs and sternum together form the rib cage.

(4) 11th and 12th pairs of ribs are called vertebrochondral ribs.

58) Consider the following four statements (a-d) and select the option which includes all the **correct** ones only:
(a) Myasthenia gravis is an auto immune disease.
(b) The progressive degeneration of skeletal muscles is called tetany.
(c) Deposition of uric acid crystals leads to osteoporosis.
(d) Inflammation of joints is known as arthritis.
Options:

(1) Statements (a), (b) and (c)

(2) Statements (a), (c) and (d)

(3) Statements (a) and (c)

(4) Statements (a) and (d)

59) Which of the following is **not a correct** match for disease and its symptoms ?

(1) Tetany - low Ca^{2+} level in blood causing rapid spasms in muscle

(2) Muscular dystrophy - Genetic disorder resulting in progressive degeneration of skeletal muscle

(3) Osteoporosis - age-related disorder characterised by increase bone mass and decreased chance of fracture.

(4) Arthritis - inflamed joints

60) Select the **incorrect** option :

(1) 8th, 9th and 10th pairs of ribs articulates directly with 7th ribs

(2) 11th and 12th pairs of ribs are connected to thoracic vertebrae.

(3) Each rib is a flat thin bone and all the ribs are connected ventrally to the thoracic vertebrae

(4) There are 7 pairs true ribs, 3 pairs of false ribs and 2 pairs of floating ribs

61) Adrenal cortex regulates .....
Thyroid gland regulates .....
Parathyroid regulates .....
Read above incomplete sentences and find out correct option to complete them :-

(1) BMR, Ca^{+2} metabolism, release of MSH

(2) Metabolism, BMR, release of few amount of sex steroids

(3) Pigmentation, growth, release of growth hormone

(4) Electrolytic balance, BMR, Ca^{+2} metabolism

62) Adrenal, Thyroid, Liver, Pancreas, Pituitary, Lungs, Thymus, Kidney, Pineal.
Select the correct option for number of hormone for organised & unorganised gland respectively.

(1) 4, 5

(2) 5, 3

(3) 6, 2

(4) 7, 2

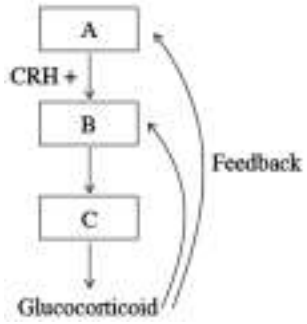
63) Select the hormone which is associated with the function like.
(a) Erythropoiesis
(b) Immunity
(c) Carbohydrate metabolism
Chose the **correct** answer from the options given below

(1) Iodothyronins

(2) Glucocorticoids

(3) Catecholanines

(4) Somatomedins



64) Identify the A,B and C in given feedback Mechanism

	A	B	C
(1)	Posterior Pituitary	Hypothalamus	Adrenal cortex
(2)	Hypothalamus	Anterior pituitary	Adernal medula
(3)	Anterior pituitary	Adrenal cortex	Hypothalamus
(4)	Hypothalamus	Anterior pituitary	Adrenal cortex

(1) 1

(2) 2

(3) 3

(4) 4

65) **Assertion:** The hormones produced by hypothalamus are of two types, the releasing hormones and the inhibiting hormones.
Reason: A hypothalmic hormone called Gonadotrophin releasing hormone(GnRH) stimulates the pituitary gland to synthesis and release of gonadotrophins. On the other hand, somatostatin from the hypothalamus inhibit the growth hormone from the pituitary.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

66) How many hormones are not secreted from pars distalis ?
GH, PRL, TSH, T_4, TCT, PTH, MSH, ADH

(1) 3

(2) 4

(3) 5

(4) 6

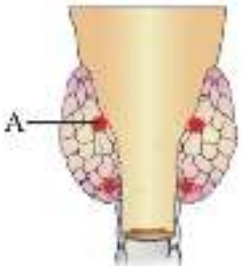
67) Which of the following endocrine glands diminishes gradually ?

(1) Thyroid

(2) Thymus

(3) Adrenal

(4) Pancreas



68) Identify the following diagram and give correct answer for A point :-

(1) It is thyroid gland. It's hormone deficiency related with defective development of growing baby during pregnancy

(2) It is adrenal medulla. It's hormones called catecholamines or hormones of emergency.

(3) It is parathyroid gland related to maintenance of Ca^{+2} in blood by bone reabsorption

(4) It is thymus gland related to immune development of body through activity of lymphocytes.

69) Read the following statements :-
(a) Hormones are nutrient chemicals.
(b) Pancreas is organized endocrine gland.
(c) Anterior pituitary is under direct neural regulation of hypothalamus.
(d) Growth hormone is secreted by anterior pituitary.
(e) Deficiency of vasopressin cause acromegaly.
Choose the **correct** options :-

(1) a,b, & c

(2) b,c & d

(3) a,c & d

(4) b & d

70) Which hormone regulate chromosome function by interaction of hormone-receptor complex with genome?

(1) FSH

(2) Thyroxine

(3) Oxytocine

(4) Epinephrine

71) Which of the following hormone decreases the blood pressure?

(1) Aldosterone

(2) ADH

(3) ANF

(4) Insulin

72) P is a small, round, reddish structure located on the dorsal side of forebrain. It releases a hormone Q which controls diurnal rhythm of the body. P and Q are

(1) Hypothalamus, MSH respectively

(2) Pineal gland, melanin respectively

(3) Pineal gland, melatonin respectively

(4) Pituitary gland, MSH respectively

73) Glucose homeostasis and Ca^{2+} homeostasis is related with _____ and _____ group of hormones in human body.

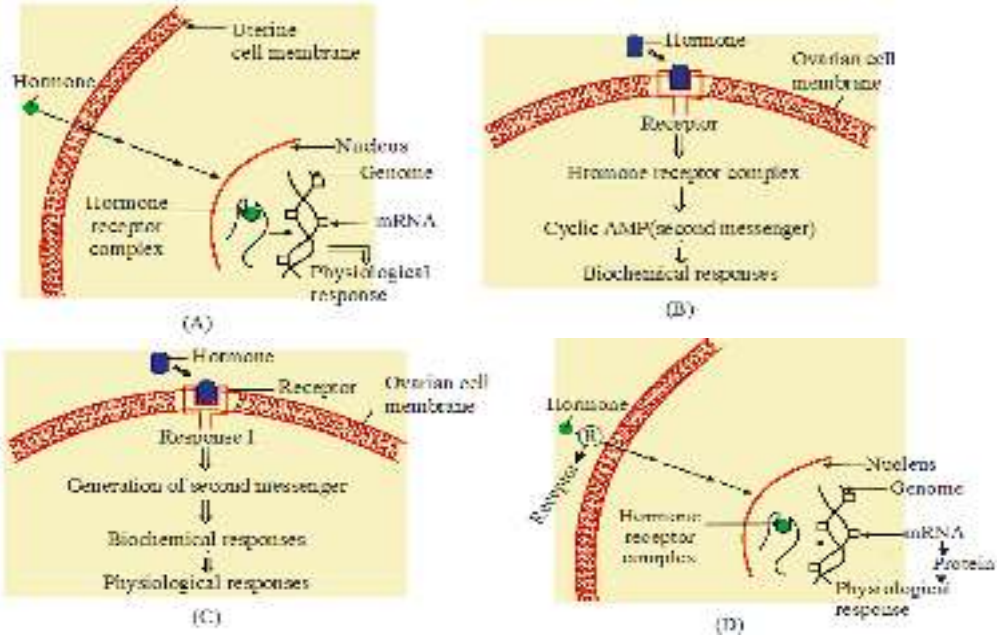
(1) Insulin – glucagon, PTH – Thymosin

(2) Insulin – Cortisol, PTH – MSH

(3) Insulin – glucagon, MSH – Melatonin

(4) Glucagon – Insulin, TCT – PTH

74) In following diagrams which shows the working of steroid hormone ideally.



(1) Fat soluble → A

(2) Water soluble → B

(3) Water soluble → C

(4) Fat soluble → D

75) Which hormone produces anti-inflammatory reactions and suppresses the immune response.

(1) Thyroxine

(2) Cortisol

(3) Parathyroid hormone

(4) Thymosine

76) Find out **correct** statement :-

(A) Somatic neural system relay impulses from CNS to skeletal muscles.

(B) ANS transmits impulses from CNS to voluntary muscles.

(C) Neural organisation is very simple in *Hydra* where as better organised in insects.

(D) Neuron can detect, receive and transmit stimuli at both sides of axon.

(1) A & B correct

(2) B & C correct

(3) C & D correct

(4) A & C correct

77)

In resting condition the outer surface of the axonal membrane possesses a ___A___ while its inner surface has ___B___ and therefore is ___C___.

	A	B	C
(1)	Positive charge	Negative charge	Depolarised
(2)	Positive charge	Negative charge	Polarised
(3)	Negative charge	Positive charge	Polarised
(4)	Positive charge	Negative charge	Hyper polarised

(1) 1

(2) 2

(3) 3

(4) 4

78) Association area of cerebral cortex are responsible for complex functions like –

(1) Intersensary association

(2) Memory

(3) Communication

(4) All of the above

79) Consider the following four **statement** (a-d) and select the option which includes the correct one only :
(a) The gap in between two unmyelinated sheath is called nodes of ranvier
(b) myelinated nerve fibres are present in spinal and carnial nerve
(c) Unmyelinated neve fibre is enclosed by a schwann cell that does not form a myelin sheath around the axon. It is commonly found in autonomous and somatic neural system
(d) Each branch of dendrite terminates put a bulb like structure called synaptic knob which possesses synaptic vesicles, which contain a chemical called neurotransmitter

(1) Statement (c) and (d)

(2) Statement (b), (c) and (d)

(3) Statement (a) and (c)

(4) Statement (a), (c) and (d)

80) Brain–stem of human brain consist of __A__, pons & medulla oblongata.
Find correct option & fill in the blank space.

(1) Crebrum

(2) Cerebellum

(3) Mid brain

(4) Diencephalon

81) Limbic system needs coordination from which of the following brain part to control sexual behaviour?

(1) Pons

(2) Corpra quadrigemina

(3) Thalamus

(4) Hypothalamus

82) Which part of brain is supposed to be damaged if in an accident, a person lost control of water balance, hunger and body temperature ?

(1) Cerebellum

(2) Hypothalamus

(3) Medula oblongata

(4) Corpora quadrigemina

83) In central nervous system myelin sheath is produced by __A__ and in Peripheral Nervous system myelin sheath is produced by __B__.
Select the **correct** option about A and B respectively :-

(1) Oligodendrocytes and Astrocytes

(2) Oligodendrocytes and Schwann cells

(3) Schwann cells and Astrocytes

(4) Schwann cells and Oligodendrocytes.

84) Choose the **correct** statement regarding synapse :-

(1) Electrical current can flow directly from one neuron into the other across the chemical synapse

(2) Electrical synapse use neurotransmitter

(3) Impulse transmission across chemical synapse is slower than that across an electrical synapse

(4) The membranes of presynaptic and postsynaptic neurons are in close proximity in an chemical synapse

85) Nissl's granules are found in ..... and helps in ..... :-

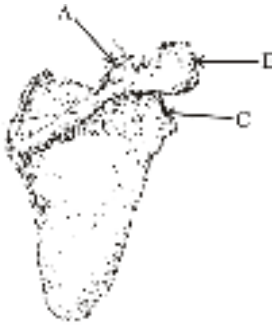
(1) Cyton only, lipid synthesis respectively

(2) Cyton and Axon, Protein synthesis respectively

(3) Dendron and Axon, Protein synthesis respectively

(4) Dendron and Cyton, Protein synthesis respectively

86) Label the following diagram of right scapula with choosing the suitable option



	A	B	C
1	Corocoid process	Acromian process	Glenoid Cavity
2	Acromian process	Corocoid process	Acetabulum cavity
3	Acromian process	Corocoid process	Glenoid cavity
4	Acromian process	Corocoid process	Acetabulum cavity

(1) 1

(2) 2

(3) 3

(4) 4

87) Following statements are true about electrical synapse except:-

(1) Gap junctions helps in bidirectional impulse transmission.

(2) Receptor proteins not involved at synapse

(3) Synaptic delay occurs

(4) Transmission of an impulse across electrical synapse is very similar to impulse conduction along a single axon.

88) Pneumotaxic centre which can moderate the functions of the respiratory rhythm centre is present at

(1) Pons region of brain

(2) Thalamus

(3) Spinal cord

(4) Right cerebral hemisphere

89) Which of the following statement is false for saltatory conduction :-

(1) Takes place in myelinated axons only

(2) Energy is conserved

(3) This conduction is rapid

(4) More number of Na^+–K^+ pump are required for this conduction

90) Which of the following is responsible for intersensory association between sensory & motor area ?

(1) Motor area
(2) Sensory area
(3) Medulla
(4) Association area

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	2	2	1	2	2	2	3	1	1	4	2	2	1	4	1	1	3	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	4	1	3	4	1	3	4	3	1	2	3	2	3	4	3	3	1	4	4	2
Q.	41	42	43	44	45															
A.	1	3	1	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	2	2	4	4	1	4	1	4	3	2	3	4	4	1	4	3	1	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	1	2	3	1	4	1	2	2	3	1	1	4	2	2	2	4	2	2
Q.	86	87	88	89	90															
A.	3	4	4	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	1	3	3	4	2	4	3	4	4	3	1	2	1	3	3	3	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	1	2	4	4	3	4	3	2	2	4	2	4	1	2	1	1	4	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	3	3	3	3	2	3	4	4	2	4	4	4	2	2	3	4	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	2	4	1	2	2	3	4	2	3	3	4	1	2	4	2	4	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	2	3	4	1	3	1	4	4										

SOLUTIONS

PHYSICS

1) $PT = C$, $PV = nRT$

$T \propto PV$

$P(PV) = C$

$P^2V = C$

$PV^{1/2} = C$

$x = \frac{1}{2}$

$C = C_V + \frac{R}{1-x}$

$4.5\ R = \frac{f}{2}R + \frac{R}{1-\frac{1}{2}}$

$2.5\ R = \frac{fR}{2}$

$f = 5$

2) Heat gain by Ice = Heat loss by water

$-20^\circ C \rightarrow 0^\circ C$ (Ice) $\quad 0^\circ C \rightarrow 10^\circ C$ (water)

$m_i\ S_i\ \Delta Q_i = m_w\ S_w\ \Delta Q_w$

$10 \times \frac{1}{2} \times 20 = 10 \times 1 \times 10$

Final condition ⇒ 0°C Ice of 10g

+

0°C water of 10g

3) Let R_A, R_B and R_C be the thermal resistance of rods A, B and C respectively. Then

$R_A = \frac{2l}{KA}$; $R_B = \frac{l}{2K.2A} = \frac{l}{4KA}$

and $R_C = \frac{l}{4K.2A} = \frac{l}{8KA}$

Since thermal current

$H = \frac{Q}{t} = \frac{\theta_1 - \theta_2}{l/KA} = \frac{\theta_1 - \theta_2}{R}$a

or $RH = \theta_1 - \theta_2$

$\therefore \quad 100 - \theta = R_AH_A$ ...(i)

$\theta - 50 = R_BH_B$ ...(ii)

$\theta - 0 = R_CH_C = R_C(H_A - H_B)$ ...(iii)

On substituting values of R_A, R_B and R_C, we get

$100 - \theta = \frac{2l}{KA}.\ H_A \Rightarrow H_A = \frac{KA}{2l}(100 - \theta)$

$\theta - 50 = \frac{l}{4KA}H_B \Rightarrow H_B = \frac{4KA}{l}(\theta - 50)$

$\theta = \frac{l}{8KA}(H_A - H_B)$

$$= \frac{1}{8KA}\left[\frac{KA}{2l}(100-\theta)-\frac{4KA}{l}(\theta-50)\right]$$

$\theta = 20°C$

4)

$VT^2 = \text{constant}$

$$\frac{\Delta V}{V}+\frac{2\Delta T}{T}=0$$

$$\frac{\Delta V}{V}=\frac{-2\Delta T}{T}$$

$$\frac{\Delta V}{\Delta TV}=\frac{-2}{T}$$

$$\gamma=\frac{-2}{T}$$

5) $V_{rms}=\sqrt{\frac{3RT}{Mw}}$

$$\frac{(V_{rms})\,He}{(V_{rms})\,H_2}=\sqrt{\frac{T_{He}}{T_{H_2}}\times\frac{M_{WH_2}}{M_{WH_e}}}$$

$$\frac{1}{\sqrt{2}}=\sqrt{\frac{T_{He}}{273}\times\frac{2}{4}}$$

$$\frac{1}{2}=\frac{T}{546}$$

$T = 273\ K = 0°C$

6) $\frac{m_1T_1}{P_1}=\frac{m_2T_2}{P_2}$

$$\frac{32\times330}{10}=\frac{m_2300}{5}$$

$$m_2=\frac{32\times330}{300\times2}=\frac{88}{5}gm$$

Leaked mass $= m_1 - m_2$

$$=32-\frac{88}{5}=\frac{72}{5}gm$$

7) $W = nR(2T^0)\ln\frac{2P_0}{4P_0}$

$$= 2RT^0\ln\frac{1}{2}$$

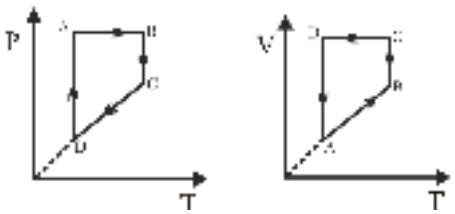
$W = -2RT_0\ln2$

 Work done on gas $= 2RT_0\ln 2$

8) For given graph.

Process AB is isobaric, BC and DA are isothermal, CD is isochoric.

If we plote P-T & V-T graph then



9) $T_m \propto \frac{1}{\lambda} \Rightarrow E \propto T^4 \Rightarrow E \propto \frac{1}{\lambda^4}$

$\frac{E_1}{E_2} = \left(\frac{\lambda_2}{\lambda_1}\right)^4$

$\frac{4}{E_2} = \left(\frac{1}{2}\right)^4 \Rightarrow E_2 = 64$

10) If no heat flow through CD then it is like balanced wheat stone bridge $K_1K_4 = K_2K_3$.

11) $T = \frac{V}{\mu R} P \Rightarrow y = mx$

As $\theta_2 > \theta_1 \Rightarrow$ Slope $= \frac{V}{\mu R}$

Slope $\propto V$

$\theta_2 > \theta_1 \Rightarrow V_2 > V_1$

12)

6K	3K	K

$R_{eq} = \frac{R}{6} + \frac{R}{3} + R$

$R_{eq} = \frac{R + 2R + 6R}{6}$

$R_{eq} = \frac{9R}{6}$

$\frac{3\ell}{A.K_{eq}} = \frac{9}{6}R = \frac{9}{6} \times \frac{\ell}{KA}$

$K_{eq} = 2K$

13) $\frac{d\theta}{dt} = -K(\theta - \theta^0)$

$\frac{\tan\phi_2}{\tan\phi_1} = \frac{-K(\theta_2 - \theta_0)}{-K(\theta_1 - \theta_0)}$

$\frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$

14) Heat required to heat 1 Lt water = $ms\Delta\theta$

$= 1 \times 4200 \times (60 - 20)$

$= 4200 \times 40$

requred time = $\frac{4200 \times 40}{1000}$
= 42 × 4 = 168
i.e. 2 min 48 second

15) $\frac{dQ}{dt} = L\frac{dm}{dt} = L\rho A\frac{d\ell}{dt}$ ...(1) $\frac{dQ}{dt} = \frac{kA}{\ell}[0 - (-\theta)]$ ...(2)
From (1) & (2)
$L\rho A\frac{d\ell}{dt} = \frac{kA}{\ell}(\theta)$
$\frac{d\ell}{dt} = \frac{k\theta}{\rho\ell L}$

16)

$Q = \mu C_v \Delta T$
$= \frac{1}{4} \times \frac{3}{2}R(T_2 - T_1)$
$= \frac{3}{8}kN_A(T_2 - T_1)$

17)

Generated by Allie

Problem Statement: The question asks how the time period of a simple pendulum changes when its temperature is increased by ΔT, considering that the pendulum's length changes due to thermal expansion.

Underlying Concept: The core concept is that the length of the pendulum changes with temperature due to linear thermal expansion, which in turn affects the pendulum's time period because the time period depends on the square root of its length.

Relevant Formulas: The time period of a simple pendulum: $T = 2\pi\sqrt{\frac{l}{g}}$ Length expansion with temperature: $l' = l(1 + \alpha\Delta T)$ New time period: $T' = 2\pi\sqrt{\frac{l'}{g}} = T\sqrt{1 + \alpha\Delta T}$

Step-by-Step Calculation: Using $\sqrt{1 + x} \approx 1 + \frac{x}{2}$ for small $x = \alpha\Delta T$, we find:

$T' \approx T\left(1 + \frac{\alpha\Delta T}{2}\right)$ Change in time period: $\Delta T_{period} = T' - T \approx \frac{1}{2}\alpha T\Delta T$

Tips and Tricks: Use the binomial approximation $\sqrt{1 + x} \approx 1 + \frac{x}{2}$ for small thermal expansions to simplify the expression.

Common Mistakes: Forgetting to apply the square root when substituting the expanded length into the time period formula or ignoring the linear approximation for small expansions.

Why Other Options Are Incorrect?: Options without the $\frac{1}{2}$ factor are incorrect because the time period depends on the square root of the length, so the change is half of the relative change in length. Answers missing the original time period t or mixing up factors of α and ΔT are incorrect.

18)

Generated by Allie

Problem Statement: The question asks which heating curve correctly shows how a block of ice, initially at -100Â°C, is heated until it becomes steam at 100Â°C. You need to identify the correct shape of the temperature vs. heat graph that represents the stages of heating, melting, boiling, and vaporization.

Underlying Concept: This problem involves the heating curve of water and its phase changes. When a solid is heated, its temperature rises until it reaches its melting point where it changes phase to liquid at constant temperature. Then the liquid heats up until it reaches its boiling point, where it vaporizes at constant temperature. Key concepts include sensible heat (temperature change), latent heat of fusion (melting), and latent heat of vaporization (boiling).

Tips and Tricks: Remember that during phase changes, temperature stays constant even as heat is added. The curve rises when the substance's phase is uniform (solid or liquid) and flattens during melting and boiling. Also note that starting below 0Â°C just shifts the initial temperature but does not change the curve's shape.

Common Mistakes: Students sometimes think temperature continuously rises through melting and boiling; in reality, temperature remains constant during phase changes. Another mistake is not recognizing multiple rising and flat segments in sequence.

Why Other Options Are Incorrect?: Graphs that do not show two flat plateaus at 0Â°C and 100Â°C are incorrect because they miss the melting or boiling phases. Curves that show continuous linear temperature increase with no flat sections do not represent phase transitions.

19) $C_P - C_v = m$, for hydrogen ($M_1 = 2$)
$C_p - C_v = n$, for nitrogen ($M_2 = 14$)
For hydrogen: $$C_P - C_V = \frac{1}{M_1}\frac{dQ}{dT} = m$$
For nitrogen: $$C_P - C_V = \frac{1}{M_2}\frac{dQ}{dT} = n$$
$$\frac{m}{n} = \frac{\frac{1}{M_1}\frac{dQ}{dT}}{\frac{1}{M_2}\frac{dQ}{dT}} = \frac{M_2}{M_1} = \frac{14}{2} = 7$$

 m = 7n

20)

On increasing temperature, peak will shift left.

21) **Question is Asking About:**
The question provides two equations representing simple harmonic motions and asks for the ratio of their anplitude.

Concept:
S.H.M.

Formula:
$A = \sqrt{a^2 + b^2}$
$y = A \sin(wt + \phi)$

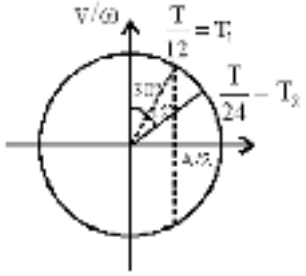
Solution/Explanation/Calculation:
$y_1 = 10\sin\left(3\pi t + \frac{\pi}{4}\right)$
$y_2 = 5\left(\sin(3\pi t) + \sqrt{3}\cos(3\pi t)\right)$
$y_2 = 5\sin(3\pi t) + 5\sqrt{3}\cos(3\pi t)$
Such that $A_1 = 10$
$A_2 = \sqrt{5^2 + \left(5\sqrt{3}\right)^2}$
$A_2 = \sqrt{25 + 75}$
$A_2 = \sqrt{100}$
$A_2 = 10$
$\frac{A_1}{A_2} = \frac{1}{1}$
1 : 1

Hence, option (4) is correct.

Level : Medium

22) $A_{net} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$
$TE = \frac{1}{2}KA^2 = \frac{1}{2} \times 100 \times \left(4\sqrt{2}\right)^2$
$= 50 \times 16 \times 2 = 1600$ J

23) equilibrium position $F_{net} = 0$ but after that spring force increases & net force becomes in upward direction therefore of max compresion acc. will be non zero.
Let compression in spring is x.
at max compression
$\frac{dx}{dt} = 0 \Rightarrow v = 0$

24) 

25) $T \propto \frac{1}{\sqrt{k}}$

$\Rightarrow T_1 : T_2 : T_3 = \frac{1}{\sqrt{k}} : \frac{1}{\sqrt{k/2}} : \frac{1}{\sqrt{2k}}$

$= 1 : \sqrt{2} : \frac{1}{\sqrt{2}}$

26) $\frac{1}{2}kA^2 = 0.06 - 0.02$

$\frac{1}{2}k\left(\frac{10}{1000}\right)^2 = 0.04$

k = 800 N/m

27)

$k_{eq} = 2k + k + 2k$

$= 5k$

$T = 2\pi\sqrt{\frac{M}{5K}}$

28) $\frac{\Delta T}{T} \times 100 = \left[\sqrt{\frac{\ell_2}{\ell_1}} - 1\right] \times 100$

$= \left[\sqrt{\frac{196}{100}} - 1\right] \times 100$

= 40%

29)

Explanation:

Explain why pendulum clocks run slower in summer and faster in winter due to thermal effects.

Concept:

This question is based on time period of a pendulum: $T = 2\pi\sqrt{(L/g)}$. If length L increases (as in summer), τ increases, so clock runs slow.

Formula:

- $T = 2\pi\sqrt{(L/g)}$

- Increase in L → Increase in T → Clock slows down

Calculation:
In summer:
Length increases → Time period increases → Clock ticks slower In winter:
Length decreases → Time period decreases → Clock ticks faster
Assertion is true and Reason is also true, and reason correctly explains the assertion.
Answer: Option (1)
Both Assertion \& Reason are true and the reason is the correct explanation.

30) The velocity-displacement relation is $x^2 + \frac{\nu^2}{\omega^2} = A^2$
which may be a circle if $\omega = 1$. and ellipse of $\omega \neq 1$.
The acceleration-velocity relation is
$\nu^2 + \frac{a^2}{\omega^2} = A^2$ which may be a circle if $\omega = 1$ and ellipse of $\omega \neq 1$.
Acceleration-displacement graph is straight and acceleration time graph is sinusoidal.

31) $y = A_0 + (A\sin\omega t + B\cos\omega t)$
$= A_0 + A\sin\omega t + B\sin\left(\omega t + \frac{\pi}{2}\right)$
Hence 2 SHMs are superimposed with phase difference of $\frac{\pi}{2}$
Amplitude $a = \sqrt{A^2 + B^2 + 2AB\cos\Delta\phi}$
As, $\Delta\phi = \frac{\pi}{2}$ Hence $a = \sqrt{A^2 + B^2}$
$y = A_0 + \sqrt{A^2 + B^2}\sin(\omega t + \phi)$
A_0 is mean position, and $\sqrt{A^2 + B^2}$ is amplitude

32) **(1) Asking About:**
We need to determine the path difference between the two given wave equations.
(2) Concept:

A. The given waves are in sinusoidal and cosine forms, and there is a phase difference ϕ.
B. A phase difference ϕ corresponds to a path difference Δx given by:

Path difference $= \frac{\lambda}{2\pi} \times$ Phase difference
(3) Formula:
$\Delta x = \frac{\lambda}{2\pi} \times \phi$
Since cos (θ) = sin $(\theta + \frac{\pi}{2})$, we rewrite y_2 as:
$y_2 = a_2\sin\left(\omega t - \frac{2\pi x}{\lambda} + \phi + \frac{\pi}{2}\right)$
This means the phase difference between y_1 and y_2 is $\phi + \frac{\pi}{2}$.
(4) Calculation / Explanation:
Using the phase difference to path difference formula:
Path difference $= \frac{\lambda}{2\pi}(\phi + \frac{\pi}{2})$

(5) Final Answer:
The correct answer is:
Option (2): $\frac{\lambda}{2\pi}(\phi+\frac{\pi}{2})$

33) **Question Explanation:** Find the intensity ratio of waves.

Concept: Wave Intensity

Solution: The intensity (I) of a wave is proportional to the square of its amplitude (A) and the square of its frequency (f). The relationship is $I \propto A^2 f^2$.First, let's find the ratios of amplitude and frequency from the graph:

A. **Amplitude Ratio**: The amplitude is the maximum displacement. $A_1=4$ and $A_2=1$. So, $\frac{A_1}{A_2}=\frac{4}{1}$.
.
B. **Frequency Ratio**: Frequency is the number of cycles per unit time. In the time wave □ completes one cycle, wave □ completes five cycles. So, $\frac{f_1}{f_2}=\frac{1}{5}$.

Now, we can find the intensity ratio:

$$\frac{I_1}{I_2}=\left(\frac{A_1}{A_2}\right)^2\left(\frac{f_1}{f_2}\right)^2=\left(\frac{4}{1}\right)^2\left(\frac{1}{5}\right)^2=16\times\frac{1}{25}=\frac{16}{25}$$

The ratio I_1:I_2 is 16:25.

Final Answer:(3)

34)

$f_1=\frac{1}{2\ell}\sqrt{\frac{T}{\mu}}$ = 450 – 375 = 75

$\lambda=\frac{1}{2\times 75}\times\sqrt{\frac{360}{4\times 10^{-3}}}=\frac{300}{150}$ = 2m

m = μ × 2 = 8×10^{-3} kg

35)

$f=\frac{V}{\lambda}=\frac{10\,cm/s}{0.5\,m}=\frac{100\,cm/s}{5\times100\,cm}$

$=\frac{1}{5}s^{-1}$

a = 10 cm

Let y = asin(kx – ωt)

V_p = –aωcos(kx – ωt)

for P, y = 5 cm

asin(kx – ωt) = 5 cm

10 cm sin(kx – ωt) = 5 cm

$\sin(kx - \omega t) = \frac{1}{2}$
$(kx - \omega t) = 150^{\circ}$
Now
$V_P = -a\omega\cos(kx - \omega t)$
$= -a(2\pi f)\cos 150^{\circ}$
$= -\frac{10}{100} \times 2\pi \times \frac{1}{5} \times -\frac{\sqrt{3}}{2}$
$= \frac{\sqrt{3}\pi}{50}$ m/s
$\vec{V}_P = \frac{\sqrt{3}\pi}{50}\hat{j}$ m/s

36) $v_s = \sqrt{\frac{\gamma RT}{M_w}} \Rightarrow v_s \propto \sqrt{T}$
$\frac{v_{s_1}}{v_{s_2}} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v_0}{v_{s_2}} = \sqrt{\frac{273}{546}} = \frac{1}{\sqrt{2}}$
$v_{s_2} = \sqrt{2}v_0 = 1.4v_0$

37) **Question Explanation :**
We need to determine the speed of sound in a gaseous mixture of 1 mole of Helium and 1 mole of Hydrogen at 127ºC, using the concept of the speed of sound in gases.

Concept :
This question is based on Speed of Sound in a Gaseous Mixture.

Formula Used :
Speed of sound in a gas:
$$v = \sqrt{\frac{\gamma RT}{M}}$$
Where, symbols have their usual meaning.

Molar mass of the mixture:
$$M = \frac{n_1 M_1 + n_2 M_2}{n_1 + n_2}$$
For a mixture, the adiabatic index is given by:
$$\gamma_{mix} = \frac{\Sigma(n_i C_{p,i})}{\Sigma(n_i C_{v,i})}$$
where C_p and C_v are specific heat capacities.

Calculation :
Given Data:
Temperature T = 127ºC = (127 + 273) K = 400K
Molar masses:
Helium (He) = 4g / mol
Hydrogen (H_2) = 2g / mol
Number of moles: 1 mole He and 1 mole H_2

Step 1: Calculate Molar Mass of the Mixture

$$\frac{(1 \times 4) + (1 \times 2)}{1 + 1} = \frac{4 + 2}{2} = 3\text{g/mol} = 3 \times 10^{-3}\text{kg/mol}$$

Step 2: Calculate γ_{mix}

For Helium (monoatomic gas):

$$C_p = \frac{5}{2}R,\ C_v = \frac{3}{2}R,\ \gamma = \frac{5}{3}$$

For Hydrogen (diatomic gas):

$$C_p = \frac{7}{2}R,\quad C_v = \frac{5}{2}R,\quad \gamma = \frac{7}{5}$$

Using the formula:

$$\gamma_{mix} = \frac{(1 \times \frac{5}{2}R) + (1 \times \frac{7}{2}R)}{(1 \times \frac{3}{2}R) + (1 \times \frac{5}{2}R)}$$

$$= \frac{\frac{5}{2}R + \frac{7}{2}R}{\frac{3}{2}R + \frac{5}{2}R} = \frac{\frac{12}{2}R}{\frac{8}{2}R} = \frac{12}{8} = 1.5$$

Step 3: Calculate Speed of Sound

Using the formula:

$$v = \sqrt{\frac{\gamma_{mix} RT}{M}}$$

Substituting values:

$$v = \sqrt{\frac{1.5 \times 8.314 \times 400}{3 \times 10^{-3}}}$$

$$= \sqrt{\frac{1.5 \times 3325.6}{3 \times 10^{-3}}}$$

$$= \sqrt{\frac{4988.4}{3 \times 10^{-3}}} = \sqrt{1662800}$$

$v \approx 1290$ m / s

Final Answer:(1)

38) Frequency of tuning fork $f_1 = 512$ Hz

Frequency of sonometer string is f_2

$f_2 = f_1 + 4$ or $f_1 - 4$

As 4 beats per second

When tension in the string is slightly increased beat frequency decreases to 2 beats per second.

when tension is increased f_2 also increases.

If $f_2 = 512 + 4 = 516$ Hz and now

f_2 increases then beat frequency would increase if $f_2 = 512 - 4 = 508$ Hz and now f_2 increases, then

beat frequency decreases so

$f_2 = 508$ Hz

39) $\frac{I_1}{I_2} = \left(\frac{a_1}{a_2}\frac{n_1}{n_2}\right)^2 = \left(\frac{4}{1} \times \frac{n}{2n}\right)^2$

$= \left(\frac{2}{1}\right)^2 = \frac{4}{1}$

40)

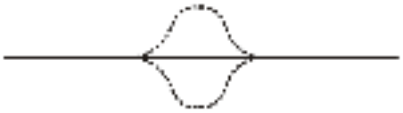
beats = $\alpha = \frac{506\pi}{2\pi} - \frac{500\pi}{2\pi}$

$\alpha = 253 - 250 = 3$

$\beta = \frac{I_{max}}{I_{min}} = \left(\frac{a_1 + a_2}{a_1 - a_2}\right)^2 = \left(\frac{4+2}{4-2}\right)^2$

$\beta = 9$

So $\beta - \alpha = 6$

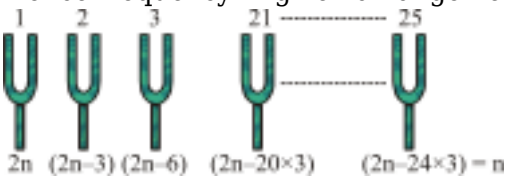
41) 

Net Displacment = 0

String becomes straight, No strain No. p.ε

42) According to the question frequencies of first and last tuning forks are 2n and n respectively.

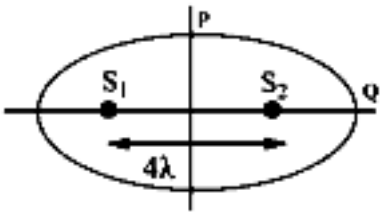
Hence frequency in given arrangement are as follows.



$\Rightarrow 2n - 24 \times 3 = n \Rightarrow n = 72$ Hz

So, frequency of 21st tuning fork

$n_{21} = (2 \times 72 - 20 \times 3) = 84$ Hz.

43) 

$S_1P - S_2P = 0$

$S_1Q - S_2Q = 4\lambda$

3 maximum lie on are between P and Q. Counting all quadrants; the total number is 12. P and Q symmetrical points are also maxima. Total number is 16

44) Let intensity due to a single person = I then $10 \log I/I_0 = 60$.

Also, Intensity due to 8 person = 8I

 Final decibel level = $10\log\left(\frac{8I}{I_0}\right) = 10\left(\log\frac{I}{I_0} + \log 8\right)$

= 60 + 10 log 8 = 60 + 30 log 2 = 60 + 30(.3010) = 69

45)

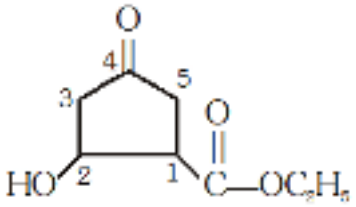
(A) PV → constant
T → constant, ΔU = 0
V increases W > 0
ΔQ > 0

(B) P ∝ T Isochoric W = 0
T increases ΔU > 0
ΔQ > 0

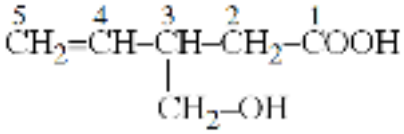
(C) P → constant
V increase W > 0
T increases ΔU > 0
ΔQ > 0

(D) V ∝ T→ Isobaric
V decreases W < 0
T decreases ΔU < 0
ΔQ < 0

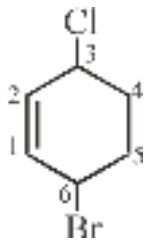
CHEMISTRY

46) 

Ethyl-2-hydroxy-4-oxo cyclo pentane carboxylate

47) 

3-Hydroxymethylpent-4-enoic acid

48) 

is correct IUPAC numbering.

49)

Conceptual

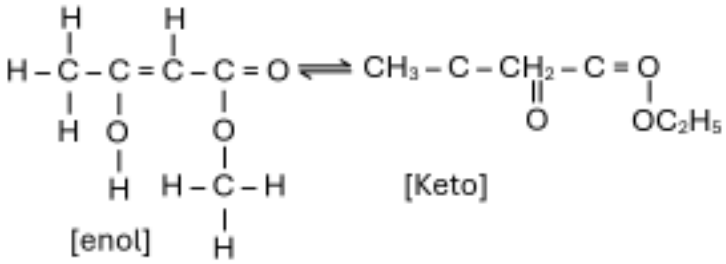
50)

Question Asking About: Concept and problems Approach
Number of sigma (σ) and pi(π) bond in enolic form of ethyl acetoacelate.

Given Data:
F

Key Concept:
(i) All single covalent bond are σ Bond.
(ii) Double covalent bond has one σ and one π bond
(iii) Triple covalent bond has one s and two π bond

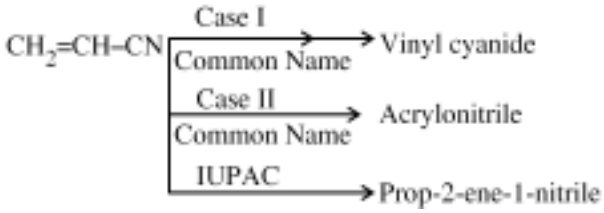
Solution/Explanation/Calculation:



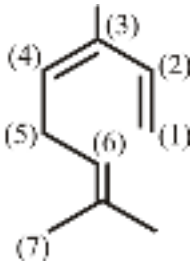
 Number of s bond in enol form is : - 18
 Number of s bond in enol form is : - 2

Conclusion:
Hence, option (4) is correct.

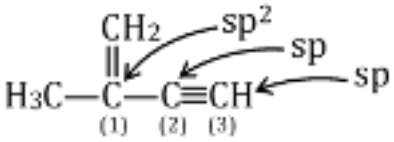
51)



52)

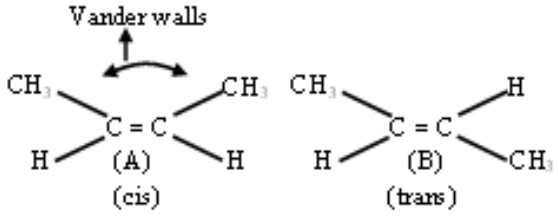


53) NCERT XI, Pg. # 343, Part – 2

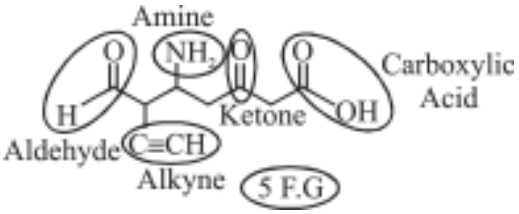
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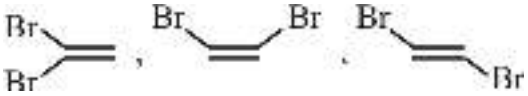
55)

Explaining the Question: Identify incorrect statement
Concept: Cis form is less stable than trans form as there are Vander walls repulsions



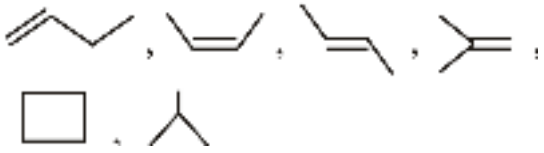
Solution:
Cis form is less stable as there is steric repulsions between the two bulky group at shorter distance Trans form has greater distance between the two $-CH_3$ groups.
Final Answer: Option_4

56) 

57) 

58)

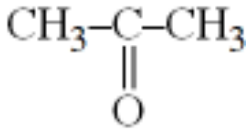
2, 2-dimethyl propane and 2-methyl butane.

59) 

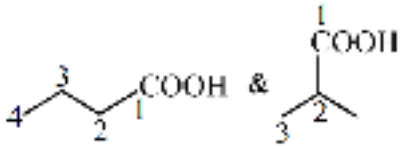
60)

Fact

61)



CH_3CH_2CHO &



62)

Both are chain isomers.

63)

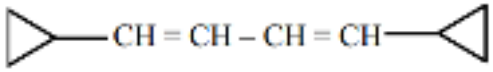
In sec. amine minimum 4C is required to show metamerism.

64)

metamerism never shown by alkenes.

65)

Explaining the Question:
Calculate total number geometrical isomers possible for given compound.
Concept: If molecule is symmetrical and number of geometrical center are even then number of geometrical isomers. = $2^{(n-1)} + 2^{\left(\frac{n}{2}-1\right)}$ where n = number of geometrical center.

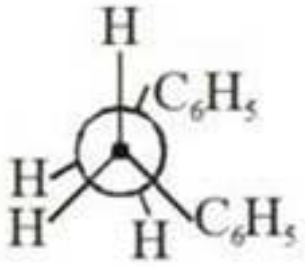


Solution:
1) Molecule is symmetrical
2) Number of geometrical center (n) = 2 [Even]
3) Number of geometrical isomers =
$\Rightarrow 2^{(n-1)} + 2^{\left(\frac{n}{2}-1\right)}$
$\Rightarrow 2^{(2-1)} + 2^{\left(\frac{2}{2}-1\right)}$
$\Rightarrow 2^1 + 2^0$
$\Rightarrow 2 + 1$
$\Rightarrow 3$
Final Answer: Option_3

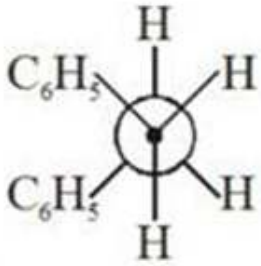
66) **Concept -**
Conformational isomers and their stability.

Answer Explanation -

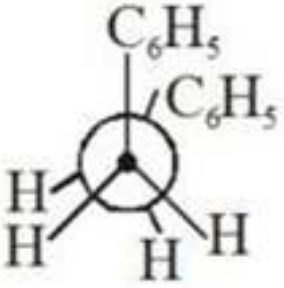
(1)



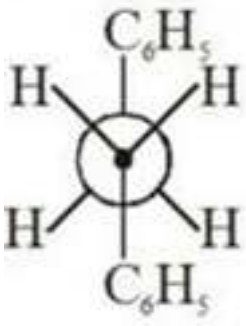
(2)



(3)



(4)



Let's analyze each Newman projection:

A. **Option 1:** The two phenyl (C6H5) groups are gauche (60° apart). This leads to significant steric strain between the large phenyl groups.

B. **Option 2:** The two phenyl (C6H5) groups are gauche (60° apart). This leads to significant steric strain between the large phenyl groups.

C. **Option 3:** The two phenyl (C6H5) groups are eclipsed (0° apart). This leads to maximum torsional and steric strain, making it highly unstable.

D. **Option 4:**The two phenyl (C6H5) groups are anti (180° apart). This minimizes steric strain between the phenyl groups.

Now, let's consider the specific interactions:

A. **Steric Strain:** Phenyl groups are large and bulky, so minimizing steric strain is crucial for stability.
B. **Torsional Strain:** Eclipsed conformations have high torsional strain.

Therefore, the most stable conformation is the one where the two phenyl groups are anti.

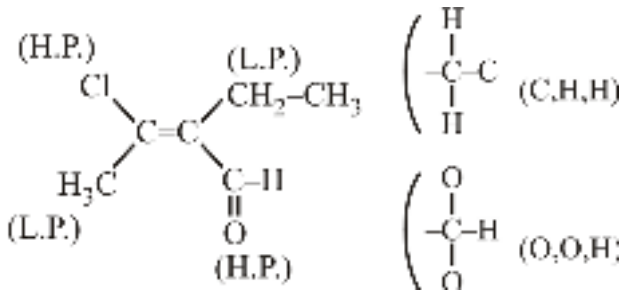
Final Answer:

Option 4

Question Level:

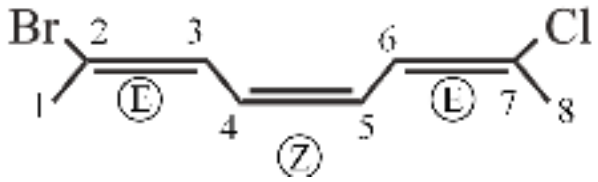
Moderate

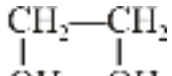
67)



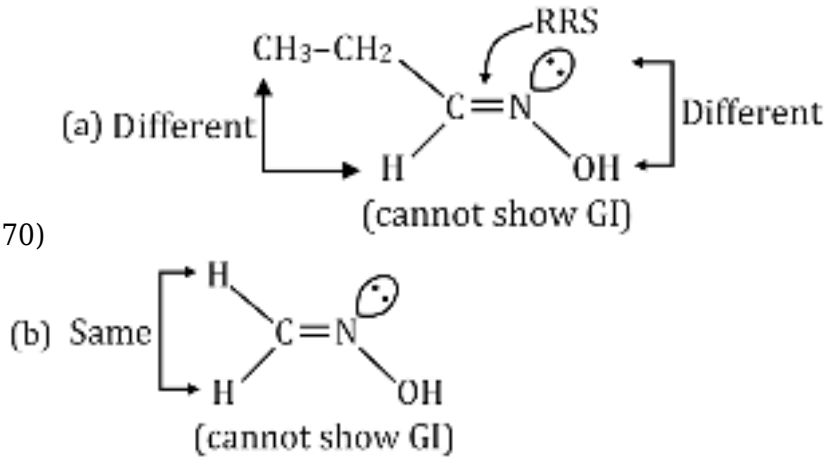
E-Isomer [According to CIP rule]

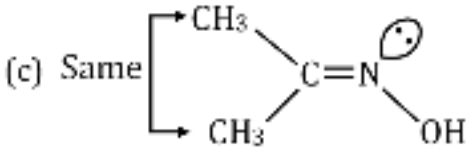
68)



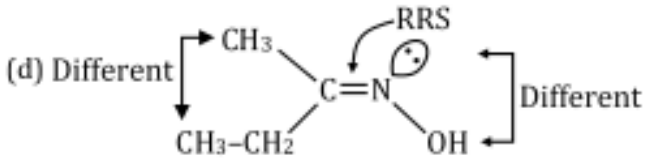
69)  can exhibit conformational isomerism.

70)





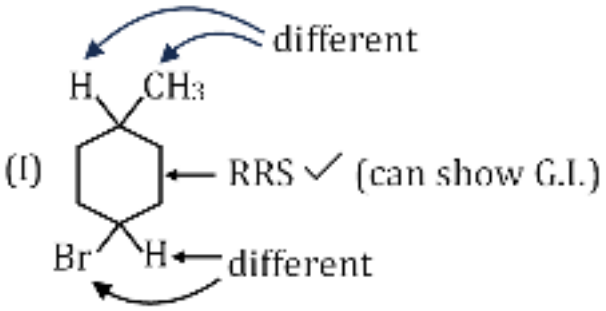
(cannot show GI)

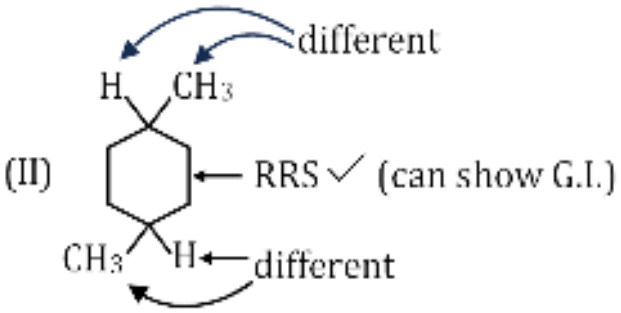


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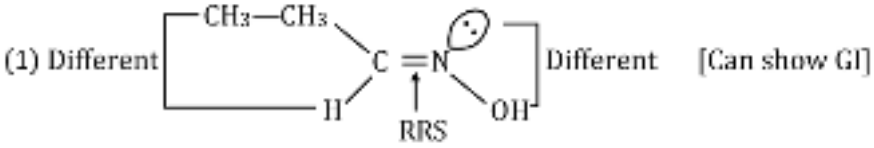
⇒ (a) & (d) can show geometrical isomerism.

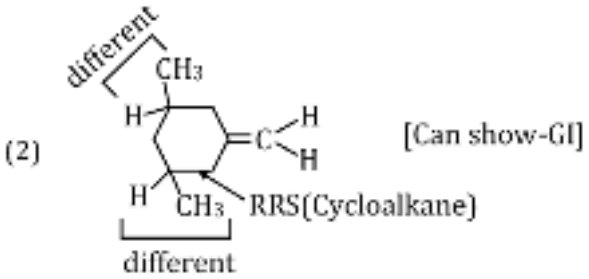
71)

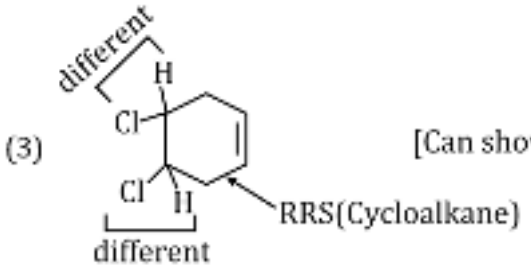




72)



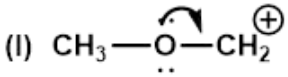


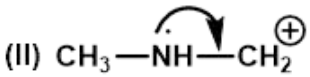
(3) 

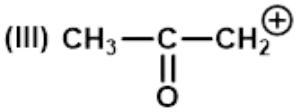
[Can show-GI]

73)

Solution/Explanation: -

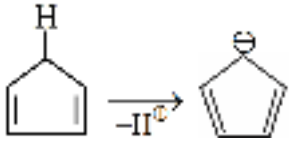
(I) 
Oxygen donate lone pair through resonance (complete octet), E.N. = N < O

(II) 
Nitrogen donate lone pair through resonance (complete octet), E.N. = N < O

(III) 
Not stabilised by resonance

 Stability order = II > I > III

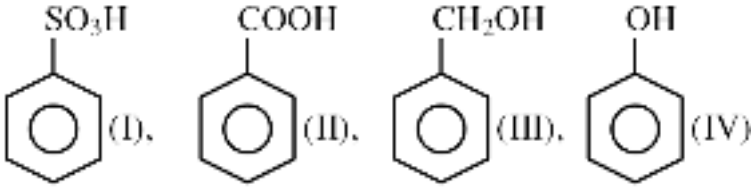
E. Final Answer: - 1

74) 

(Aromatic more stable)

75)

Explaining the Question :- The question asks to arrange the given molecules in correct order of acidic strength



Concept:- This question is based on Acidity of organic acids
Solution :- Sulfonic acid is one of the strongest organic acids-its conjugate base (sultanate) is highly stabilized by resonance and electronegative oxygen atoms.
Carboxylic acid is next, stabilized by resonance between two oxygen atoms.
Phenol (–OH) is weaker, but still acidic due to delocalization of the negative charge over the aromatic ring.
Hydroxymethyl is the weakest-it's more like an alcohol and lacks resonance or inductive stabilization.
Thus order is I > II > IV> III.
Final Answer :- (2)

76)

Question Explanation: Ordering the stability.
Concept: Stability of Carbocations
Solution: The stability of a carbocation is enhanced by resonance and the inductive effect. The compound in option I, a tertiary carbocation, is the most stable. The compound in option III, a secondary carbocation, is the next most stable. The compound in option II, a primary carbocation, is the least stable. The correct order is I > III > II. **Calculation**: N/A **Final Answer**: 3

77)

Identifying the correct statement.
Concept: Stability, Acidity, and Basicity
Solution: The statement in option A is correct. The tertiary carbocation is more stable than the primary carbocation. The statement in option B is incorrect. The statement in option C is incorrect. The statement in option D is incorrect.
Calculation: N/A **Final Answer**: 1

78)

NCERT XI Part-II, Pg # 269
Fact based

79)

Check ortho effect then –M & –I effect.

80)

B.L. ∝ more resonance

81) In  , double bond at bridgehead carbon is highly unstable, therefore will not undergo hyperconjugation.
Question Explanation: Identifying where hyperconjugation is not possible.
Concept: Hyperconjugation
Solution: Hyperconjugation requires the presence of an alpha-hydrogen atom adjacent to an unsaturated system. The compound in option D, with a positive charge on a saturated carbon atom, is not a conjugated system, so hyperconjugation is not possible. **Calculation**: N/A **Final Answer**: 4

82) N(2) is most basic as its lone pair is not delocalized.

83) Answer (4)

$\overset{+}{N}H_4$ cannot accept electrons.

84)

Stability ∝ No. of resonating structure.

85) $B.S. \propto K_b \propto \frac{+M}{-M} \propto \frac{+H}{-H} \propto \frac{+I}{-I}$

86)

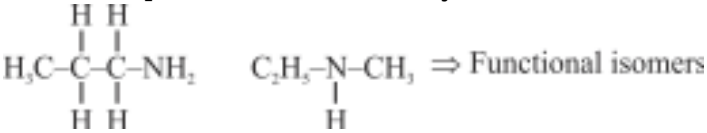
Question Explanation: Identifying the cation involved in resonance.
Concept: Resonance
Solution: Resonance requires a conjugated system. The compound in option A has a positive charge adjacent to a double bond, so it is involved in resonance. The compound in option B has a positive charge on the nitrogen atom, so the lone pair is not available for resonance.

87) **Question Explanation**: Identifying the compound that shows tautomerism.
Concept: Tautomerism
Solution: Tautomerism requires the presence of an alpha-hydrogen atom. The compound in option D, with a carbonyl group and alpha-hydrogens, can exhibit tautomerism.

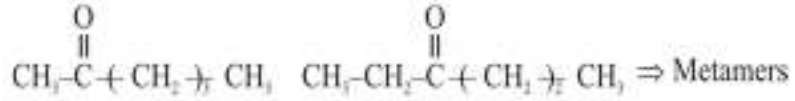
88) **Question Explanation**: Evaluating the stability of tautomers.
Concept: Tautomerism
Solution: The assertion is true: the enol form is more stable than the keto form. The reason is also true: the enol form is stabilized by aromaticity. The reason is the correct explanation of the assertion.
Enol tautomer is more stable than keto tautomer due to aromatic nature of enol.

89) **Question Explanation**: Ordering the basic strength.
Concept: Basicity
Solution: The basicity of an amine is determined by the availability of the lone pair on the nitrogen atom. Guanidine is a very strong base. Benzylamine is a strong base. The compound in option I is the most basic. The correct order is I > IV > II > III. **Calculation**: N/A
Final Answer: 3

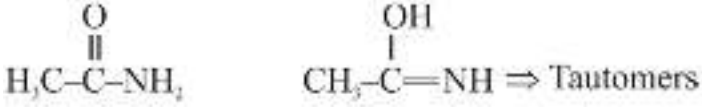
90) **Question Explanation**: Matching isomeric pairs.
Concept: Isomerism
Solution: Propanamine and N-methylethanamine are functional isomers. Hexan-2-one and Hexan-3-one are positional isomers. Ethanamide and hydroxyethanimine are tautomers. o-nitrophenol and p-nitrophenol are positional isomers.
A. Propanamine N–Methylethanamine



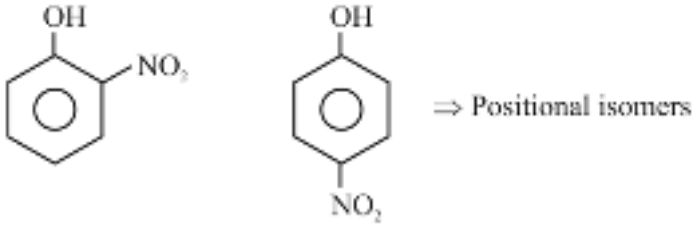
B. Hexan–2–one Hexan–3–one



C. Ethanamide Hydroxyethanimine



D. o–Nitrophenol p–nitrophenol



BIOLOGY

91)

Explain Question: The question asks how many of the given statements about cellular respiration and its pathways are factually correct.
Concept: Glycolysis/ Control of Glycolysis/ Fate of Pyruvic acid/ Aerobic respiration/Link Reaction/ TCA cycle/ Kreb's Cycle ETS/ Oxidative phosphorylation
Solution:
(a) Net gain of 2 ATP directly in glycolysis.
(b) Final hydrogen acceptor in mitochondrial ETS is oxygen.
(c) Glycolysis occurs in cytoplasm.
(d) TCA cycle of prokaryotes occurs in cytoplasm.
All are Correct.

Final answer: 4.

92)

NCERT-XI, Pg. # 159

93)

NCERT (XI) Pg # 2347

94)

NCERT Based

95)

Pg. No. 154 - XI NCERT

96) The main purpose of cellular respiration is to get energy that is utilized for functioning various purposes. Energy from chemical bonds of glucose is transferred to ATP molecules.

97)

NCERT-XI Pg. # 232

98)

New Ncert page 156.

99) NCERT Pg # 155, 158, 160

100) NCERT_234_2021-22

101) NCERT Pg. # 233

102)

NCERT pg.no 229

103)

NCERT-XI, Pg. # 159

104) NCERT XI, Page # 230

105)

NCERT Based

106) NCERT Page No. 156, 159

107) NCERT XI Page No. # 227

108) **Explanation:** PGAL (phosphoglyceraldehyde), also known as G3P (glyceraldehyde-3-phosphate), is an intermediate in glycolysis and can enter the aerobic respiration pathway. For each molecule of PGAL: Glycolysis produces 2 NADH and 2 ATP. The resulting pyruvate is converted into acetyl-CoA, which enters the Krebs cycle, generating more ATP equivalents via NADH, $FADH_2$, and GTP. After considering the total ATP generated from the glycolysis, pyruvate oxidation, and Krebs cycle, 19 or 20 ATP are produced per molecule of PGAL.
Correct Answer: 3. 19 or 20 ATP

109)

NCERT Reference: Class XI, Page No. 235

110) **Explanation:**

A. **In aerobic respiration, one glucose molecule yields 38 ATP through glycolysis, the Krebs cycle, and oxidative phosphorylation.**

B. **However, in anaerobic respiration (lactic acid fermentation), pyruvic acid is converted into lactic acid without ATP production beyond glycolysis.**

C. **Glycolysis initially produces 8 ATP (net gain of 2 ATP per glucose molecule after ATP utilization).**

D. **However, if respiration remains anaerobic, the potential 6 ATP from $NADH_2$ is lost since $NADH_2$ does not enter the electron transport system..**

Correct Answer: 3. 6 ATP is lost

111) **Explanation:**
Respiratory Quotient (R.Q.) is defined as:

Correct Answer:2. CO_2 / O_2

112) When tripalmitin is the respiratory substrate, **the amount of CO_2 released is less than the O_2 consumed.**
This is because fats, like tripalmitin, require more oxygen for complete oxidation compared to carbohydrates. This results in a Respiratory Quotient (RQ) value of less than 1, specifically around 0.7 for fats.

113) NCERT Page No. 227

114)

NCERT Pg # 231, 233

115) Carbohydrates, fats and proteins, all can be used as a substrate in cellular respiration. All of them first get converted to acetyl Co-A to enter into Krebs' cycle of aerobic cellular respiration. Thus, it is the common factor of respiration entering Krebs' cycle after breakdown of carbohydrates, fats and proteins.

116)

The assertion is true as plants have meristems that allow for continuous growth throughout their life. However, the reason is false because meristematic cells retain the capacity to divide and do not lose this ability after a certain number of divisions. Instead, the cells that are produced by the meristems soon lose the capacity to divide and differentiate into various tissues, contributing to the plant body. Therefore, A is true, but R is false.

117) NCERT XI, Pg. # 241

118)

NCERT pg.no 246

119) NCERT Page no. 241

120)

Question Explanation:

Identifying characteristics of a growth phase.

Concept:

Phases of Growth

Solution:

The Elongation phase (or zone of elongation) follows the meristematic phase. Cells in this zone increase greatly in size due to increased vacuolation (water uptake and turgor). This large cell enlargement is accompanied by new cell wall deposition to maintain structural integrity.

Final Answer:

Elongation

121) NCERT (XIth) Pg. # 243

122)

1. Auxin

• Auxin was first discovered through experiments involving the growth of Avena (oats). The "Avena curvature" bioassay (where oat coleoptiles responded to auxin) is a classic experiment associated with the discovery of auxins.
• Correct match: (a) Auxin - (iv) Avena

2. Gibberellin
• Gibberellin was discovered in Rice, specifically when it was found that rice

seedlings infected by the fungus Gibbere/la fujikuroi exhibited abnormal growth, leading to the identification of gibberellin as the growth hormone.
• Correct match: (b) Gibberellin - (ii) Rice

3. Cytokinin
• Cytokinin was first identified in Tobacco, particularly in the context of tissue culture experiments, where the hormone promoted cell division.
• Correct match: (c) Cytokinin - (i) Tobacco

4. Ethylene
• Ethylene was discovered in connection with the ripening of fruits, and the discovery was most notably linked with Oranges, as the ripening effect of ethylene was well observed in citrus fruits like oranges.
• Correct match: {d) Ethylene - (iii) Orange

123)

NCERT PAGE NO. 174

124)

Ethylene is a plant growth promoter as well as a plant growth inhibitor. Ethylene is produced by tissues of plants undergoing senescence and ripening of fruits in the form of gas.

125) NCERT Pg. # 249

126) NCERT Page no. 250

127)

NCERT Based

128) Only c

129)

NCERT Page no. 248

130) Seed dormancy is defined as a state in which seeds are prevented from germinating even under environmental conditions, normally favourable for germination. These conditions are a complex combination of water, light, temperature, gasses, mechanical restrictions, seed coats, and hormone structures.

131)

Both (A) and (R) are correct and (R) is the correct explanation of (A)Both (A) and (R) are correct and (R) is the correct explanation of (A)

132) NCERT (XI) Pg. # 229

133)

Module (E) Pg. # 208, (H) Pg. # 227

134)

NCERT Page # 216, 217, 218, 232

135) NCERT XI, Pg # 232

136) NCERT Pg. # 105

137)

The correct answer is: 2. 3 cranial bones.

Cranial bones in the list: are only 3 (Sphenoid bone, Ethmoid bone, Occipital bone)

138)

NCERT-XI Pg#227

139)

NCERT Pg. No. 218

140) NCERT Pg. # 306

141) NCERT XI Pg.# 285

142) NCERT Pg. # 306, 309 (Fig. 20.5)

143) NCERT Pg. # 307

144) NCERT Pg. # 307

145) NCERT Pg. # 305, 307 (Fig 20.2 & 20.5)

146)

NCERT Pg. # 221, 222

147)

Thoracic vertebrae, ribs and sternum together form the rib cage.

148)

NCERT XI Pg.# 227

149) NCERT Pg.# 312

150) NCERT Pg.#225

151) **Explain Question:**
Question: Complete the sentences:
Adrenal cortex regulates .....
Thyroid gland regulates .....
Parathyroid regulates .....

Concept:
This question is based on Functions of endocrine glands.

Explanation :
Adrenal cortex → regulates electrolytic balance (via mineralocorticoids like aldosterone)
Thyroid gland → regulates BMR (basal metabolic rate via thyroxine)
Parathyroid → regulates Ca^{2+} metabolism (via parathormone)

Final answer : 4

152) NCERT Pg. # 339

153) NCERT Pg. # 336

154)

NCERT Pg. # 334

155)

NCERT XII Page No. # 332

156) NCERT Pg # 332

157) NCERT Pg. # 334

158) NCERT XI pg.# 333

159) NCERT Pg. # 239,240,241

160) NCERT Pg. # 339

161)

NCERT Page No. 339

162)

NCERT-XI, Pg.# 242

163) NCERT Page # 335, 334

164)

NCERT XI Pg # 339, Fig. 22.5

165) NCERT, Pg. # 245

166) NCERT-XIth Pg # 316 (E), 316 (H)

167) NCERT (E) Pg. # 232

168) NCERT Pg. # 321

169)

Ans. Bonus

170) The **brainstem** is a key part of the brain that connects the cerebrum to the spinal cord. It is made up of three main structures: the **midbrain**, the **pons**, and the **medulla oblongata**. These parts are crucial for regulating vital involuntary functions.

171) NCERT Pg. # 321

172)

NCERT Pg. # 321

173) NCERT XII Pg # 232 (E)

174) NCERT XI Pg. # 234

175) **Explain Question :**
Nissl's granules are found in ... and help in ...

Concept :
This question is based on Structure of neuron.

Explanation :
Nissl's granules occur in cyton and dendrites (not in axon) and are involved in protein synthesis.

Final answer : 4

176) NCERT XI Pg.# 283,284

177) NCERT, Pg # 319

178)

SOLUTION/EXPLAINATION:

The **pneumotaxic center** is located in the **pons** region of the brainstem. It plays an important role in regulating the rate and pattern of breathing by modulating the activity of the respiratory rhythm center, which is located in the medulla oblongata. The pneumotaxic center helps to fine-tune and control the duration of inhalation and exhalation, influencing the overall respiratory rate.

Here’s why the other options are incorrect:
- **Thalamus**: The thalamus is involved in sensory processing and relay but is not directly responsible for regulating breathing.
- **Spinal cord**: The spinal cord transmits signals from the brain to muscles, including those for respiration, but it does not directly control respiratory rhythm.
- **Right cerebral hemisphere**: While the cerebral hemispheres are involved in higher functions, such as voluntary control over respiration, the regulation of respiratory rhythm is controlled by structures in the brainstem, not directly by the right hemisphere.

Thus, the **pons** is the correct location for the pneumotaxic center.

179)

NCERT-XI, Pg. # 232

180)

NCERT XI, Pg # 321